

How to Find Research Topics & Develop Research Questions

Vinh Dinh Nguyen
PhD in Computer Science

Outline

- **Research Topics: Thing to Know**
- **How to Find a Good Research Topic**
- **How to Develop Research Questions**
- **Sample Research Topics in Computer Vision**
- **Assignment**

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- **Assignment**

PREVIOUS



How to read a scientific research paper
How to summary a scientific research paper

TODAY



How to find a good research topic
How to develop good research questions

How to read a Research Paper

1 Abstract

Why you should read this paper

2 Introduction

High-level overview of the algorithm

3 Related Work

What other researchers did in this field

4 Approach

Detailed description of the algorithm



5 Experiments

Training, evaluation, visualization, comparisons with other papers

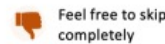
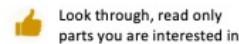
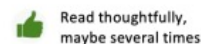
6 Conclusion

Sometimes it's about next research topics

7 References

Links to all papers that ideas were used

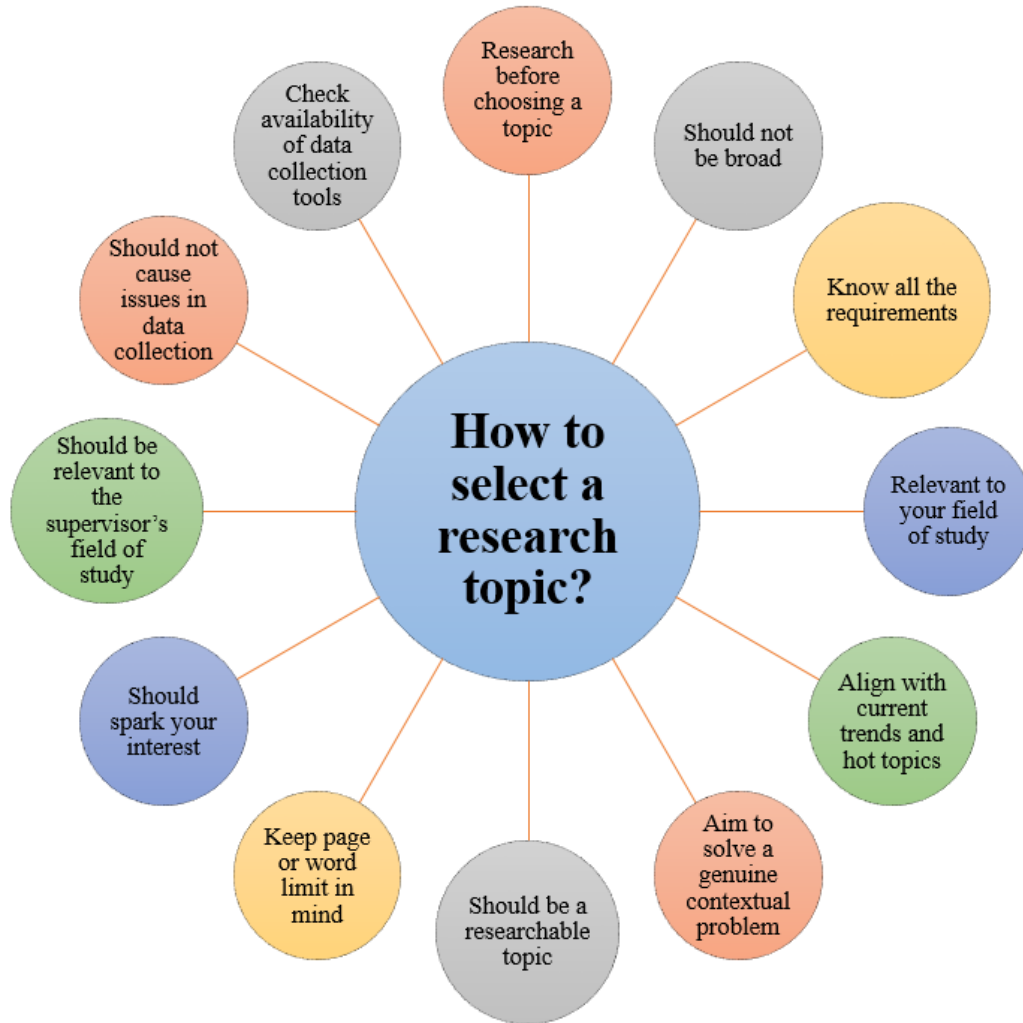
1. Abstract
2. Conclusion
3. Introduction
4. Experiments
5. The proposed method
6. Related Work



How to Summarize a Research Paper

1. Begin your summary with a brief introduction that includes the paper's title, authors, and publication information.
2. In a few sentences, introduce the research question or problem the paper addresses
3. Provide a concise overview of the methodology or approach used in the study, including the data sources, data collection and analysis methods, and any statistical techniques
4. Highlight the primary findings or results of the research. Use clear and straightforward language to communicate these findings. If the paper includes figures, tables, or graphs, refer to them as needed.
5. Conclude your summary with a closing statement that provides an overall assessment of the paper, such as its significance, relevance, or potential for future research

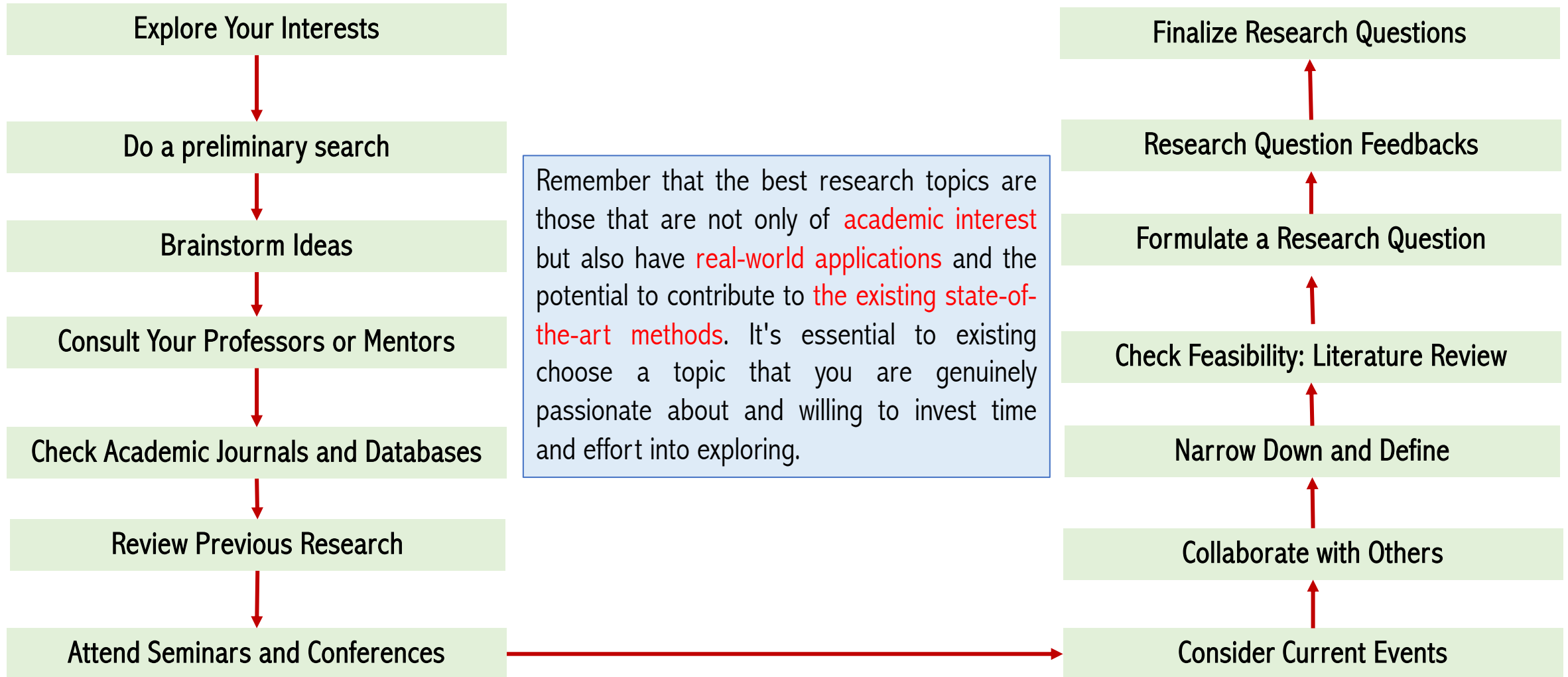
Research Topics: Thing to Know



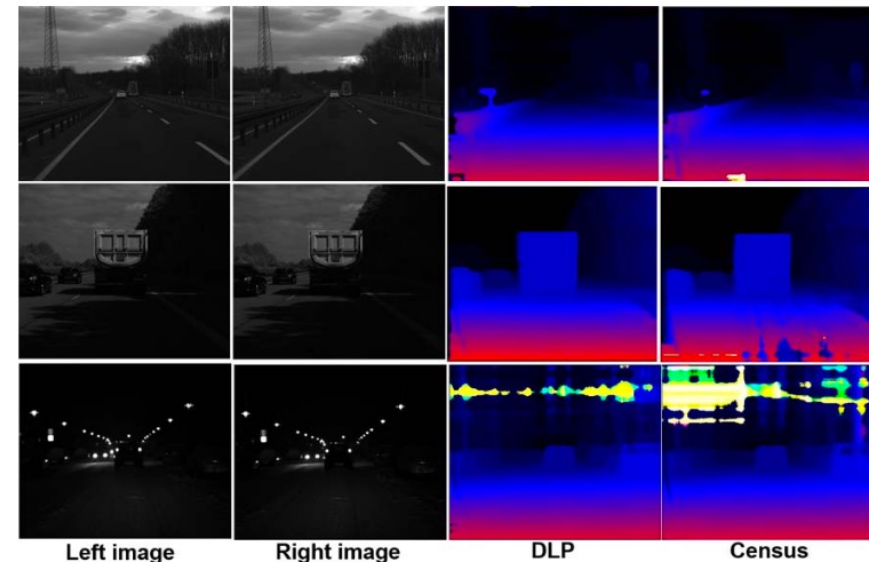
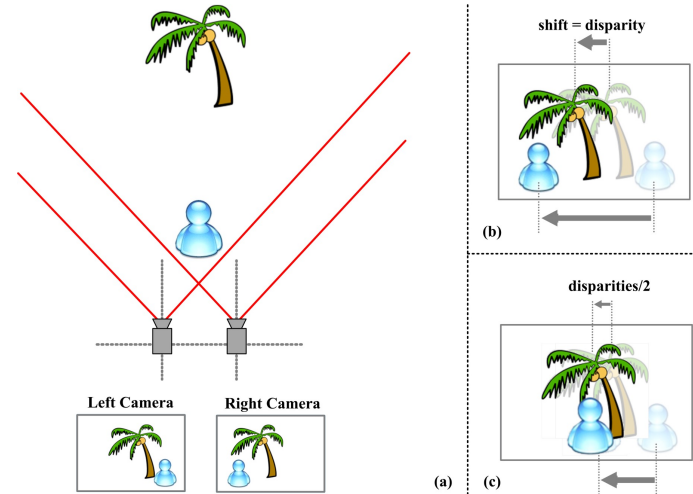
Outline

- **What is Research Topics**
- **How to Find a Good Research Topic**
- **How to Develop Research Questions**
- **Sample Research Topics in Computer Vision**
- **Assignment**

How to find Research Topics



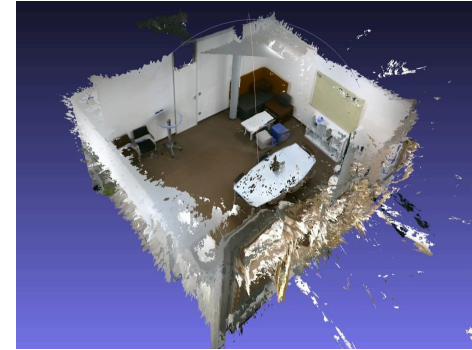
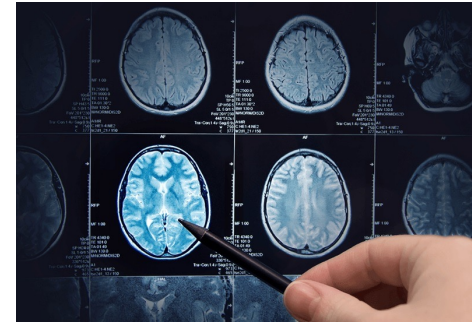
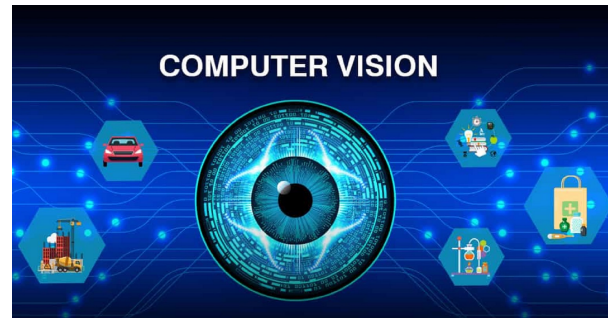
Case study 1: Stereo Matching



How to find Research Topics

Explore Your Interests

What topics fascinate you? What questions pique your curiosity? Research is more enjoyable and productive when it aligns with your interests.



SURVEILLANCE DRONES

Conduct aerial land surveys and monitor sites for security threats

WEARABLE TECH

Learns workers' habits and schedules to personalize safety alerts

3D PRINTERS

Detect and repair structural design flaws in printed construction components

AUTONOMOUS EQUIPMENT

Performs heavy-duty tasks that are dangerous or unreachable for humans

How to find Research Topics

Do a preliminary search

Read books, articles, and blogs related to your field or general interests. This will help you identify gaps in current knowledge, controversial issues, or emerging trends.

A Survey on Deep Learning Techniques for Stereo-Based Depth Estimation
Publisher: IEEE Cite This PDF

Hamid Laga ; Laurent Valentin Jospin ; Farid Boussaid ; Mohammed Bennamoun All Authors

56 Cites in Papers 5887 Full Text Views

Abstract
Document Sections
1. Introduction
2. Scope and Taxonomy
3. Database

Abstract:
Estimating depth from RGB images is a long-standing ill-posed problem, which has been explored for decades by the computer vision, graphics, and machine learning communities. Among the existing techniques, stereo matching remains one of the most widely used in the literature due to its strong connection to the human binocular system. Traditionally, stereo-based depth estimation has been addressed through matching hand-crafted features across multiple images. Despite the extensive amount of research, these traditional techniques still suffer in the presence of highly textured areas, large uniform regions, and occlusions. Motivated by their growing success in solving various

A Review of Vehicle Detection Techniques for Intelligent Vehicles
Publisher: IEEE Cite This PDF

Zhangyu Wang ; Jun Zhan ; Chunguang Duan ; Xin Guan ; Pingping Lu ; Kai Yang All Authors

28 Cites in Papers 2963 Full Text Views

Abstract
Document Sections
I. Introduction
II. Vehicle Detection Method Based on Machine

Abstract:
Robust and efficient vehicle detection is an important task of environment perception of intelligent vehicles, which directly affects the behavior decision-making and motion planning of intelligent vehicles. Due to the rapid development of sensor and computer technology, the algorithm and technology of vehicle detection have been updated rapidly. But, there are few reviews on vehicle detection of intelligent vehicles, especially covering all kinds of sensors and algorithms in recent years. This article presents a comprehensive review of vehicle detection approaches and their applications in intelligent vehicle systems to analyze the development of vehicle detection, with a specific focus on sensor types and algorithm classification. First, more than 300 research

Deep Learning for Diabetes: A Systematic Review
Publisher: IEEE Cite This PDF

Taiyu Zhu ; Kezhi Li ; Pau Herrero ; Pantelis Georgiou All Authors

70 Cites in Papers 4228 Full Text Views

Abstract
Document Sections
I. Introduction
II. Deep Learning Overview
III. Methodology

Abstract:
Diabetes is a chronic metabolic disorder that affects an estimated 463 million people worldwide. Aiming to improve the treatment of people with diabetes, digital health has been widely adopted in recent years and generated a huge amount of data that could be used for further management of this chronic disease. Taking advantage of this, approaches that use artificial intelligence and specifically deep learning, an emerging type of machine learning, have been widely adopted with promising results. In this paper, we present a comprehensive review of the applications of deep learning within the field of diabetes. We conducted a systematic literature search and identified three main areas that use this approach: diagnosis of diabetes, glucose management, and diagnosis of

Deep Generative Models for Synthetic Data: A Survey
Publisher: IEEE Cite This PDF

Peter Eigenschink ; Thomas Reutterer ; Stefan Vamosi ; Ralf Vamosi ; Chang Sun ; Klau... All Authors

1 Cites in Paper 1029 Full Text Views

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Abstract
Document Sections
I. Introduction

Abstract:
A growing interest in synthetic data has stimulated the development and advancement of a large variety of deep generative models for a wide range of applications. However, as this research has progressed, its streams have become more specialized and disconnected from one another. This is why models for synthesizing text data for natural language processing cannot readily be compared to models for synthesizing health

How to find Research Topics

Do a preliminary search

Read books, articles, and blogs related to your field or general interests. This will help you identify gaps in current knowledge, controversial issues, or emerging trends.

Andrew Ng About Publications Projects Courses Data-centric AI Contact

Dr. Andrew Ng is a globally recognized leader in AI (Artificial Intelligence). He is Founder of [DeepLearning.AI](#), Founder & CEO of [Landing.AI](#), General Partner at [AI Fund](#), Chairman and Co-Founder of [Coursera](#) and an Adjunct Professor at Stanford University's Computer Science Department.

As a pioneer in machine learning and online education, Dr. Ng has changed countless lives through his work in AI, and has authored or co-authored over 200 research papers in machine learning, robotics and related fields. In 2013, he was named to the Time 100 list of the most influential persons in the world.

[Learn more →](#)



Computer Vision Lab
Seoul National University

Home News **Members** Publication

Faculty



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[Google Scholar Profile](#)

FACULTY

Yann LeCun

• Professor

ELECTRICAL AND COMPUTER ENGINEERING

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Yann LeCun is Director of AI Research at Facebook, and Silver Professor of Data Science, Computer Science, Neural Science, and Electrical Engineering at New York University, affiliated with the NYU Center for Data Science, the Courant Institute of Mathematical Science, the Center for Neural Science, and the Electrical and Computer Engineering Department.

Research Interests
AI, machine learning, computer vision, robotics, image compression



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Geoffrey Hinton

Chief Scientific Advisor

[Website](#)

Emeritus Distinguished Professor of Computer Science, University of Toronto

Companion of the Order of Canada

Recipient ACM A.M. Turing Award

How to find Research Topics

Brainstorm Ideas

Take some time to brainstorm ideas. Write down any questions or topics that come to mind. Don't censor yourself at this stage. **Quantity over quality is key**



Stereoreo Matching

Object Detection

Object Tracking

Text-Image Retrieval

Generative AI

Graph Neural Network

Reinforcement learning

...

How to find Research Topics

Consult Your Professors or Mentors

Ask help from your professors, mentors, or advisors. They can provide valuable insights, suggest potential research areas, or help you refine your ideas.



Stereoreo Matching

Object Detection

Object Tracking

Text-Image Retrieval

Generative AI

Graph Neural Network

Reinforcement learning

...

How to find Research Topics

Check Academic Journals and Databases

Explore academic journals, databases (like IEEE, Springer or Google Scholar), and conference proceedings in your field. This will help you discover recent research and identify areas where further study is needed.



Journals & Magazines > IEEE Transactions on Pattern ... > Volume: 44 Issue: 9

On the Confidence of Stereo Matching in a Deep-Learning Era: A Quantitative Evaluation

Publisher: IEEE [Cite This](#) [PDF](#)

Matteo Poggi; Seungryong Kim; Fabio Tosi; Sunok Kim; Filippo Aleotti; Dongbo ... [All Authors](#)

10 Cites in Papers 837 Full Text Views

Abstract

Document Sections

Abstract:

Stereo matching is one of the most popular techniques to estimate dense depth maps by finding the disparity between matching pixels on two, synchronized and rectified images. Alongside with the development of more accurate algorithms, the research community

IEEE Transactions on Pattern Analysis and Machine Intelligence



CVPR CVPR My Stuff

Select Year: (2023) Dates Submit Organizers Attend Expo Media Conference Site

The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2023

Sun Jun 18th through Thu the 22nd at the Vancouver Convention Center

The IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR) is the premier annual computer vision event comprising the main conference and several co-located workshops and short courses. This year, CVPR will be single track such that everyone (with full passport registration) can attend everything. The focus will be on few selected plenary talks, scientific discussions at poster sessions and plenty of time for networking and socializing. Each paper will be presented at a poster session. All paper award candidates will have a plenary oral talk. Every attendee will have access to a personalized digital program. The virtual platform will host videos, posters and a chat room for every paper. All plenary events will be streamed!

Vancouver Canada

Google Scholar stereo matching deep learning

Articles About 98,900 results (0.08 sec)

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Since 2023

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Since 2019

Custom range...

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Review articles

[HTML] Review of stereo matching algorithms based on deep learning

K Zhou, X Meng, B Cheng - Computational intelligence and ..., 2020 - hindawi.com

... Stereo vision is a flourishing field, attracting the attention of ... the development of deep learning, stereo matching algorithms ... different stereo matching algorithms based on deep learning. ...

☆ Save Cite Cited by 59 Related articles All 11 versions

Efficient deep learning for stereo matching

W Luo, AG Schwing, R Urtasun - Proceedings of the IEEE ..., 2016 - cv-foundation.org

... stereo matching as inference in several different Markov random fields (MRFs), with the goal of smoothing the matching ... aggregation, semiglobal block matching as well as the slanted ...

☆ Save Cite Cited by 869 Related articles All 17 versions

How to find Research Topics

Review Previous Research

Look for gaps or unresolved issues in existing research. You can build upon or challenge previous findings to create a novel research project



Digging Into Uncertainty-Based Pseudo-Label for Robust Stereo Matching

Publisher: IEEE

[Cite This](#)

[PDF](#)

Zhelun Shen ; Xibin Song ; Yuchao Dai ; Dingfu Zhou ; Zhibo Rao ; Liangjun Zhan... [All Authors](#)

1

Cites in
Paper

60

Full
Text Views



Abstract

Document
Sections

- I. Introduction
- II. Related Work
- III. Our Approach

Abstract:

Due to the domain differences and unbalanced disparity distribution across multiple datasets, current stereo matching approaches are commonly limited to a specific dataset and generalize poorly to others. Such domain shift issue is usually addressed by substantial adaptation on costly target-domain ground-truth data, which cannot be easily obtained in practical settings. In this paper, we propose to dig into uncertainty estimation for robust stereo matching. Specifically, to balance the disparity distribution, we employ a pixel-level uncertainty-aware loss to expand the searching space, in this way, the unlikely

IEEE Transactions on Pattern Analysis and
Machine Intelligence

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Rethinking Training Strategy in Stereo Matching

Publisher: IEEE

[Cite This](#)

[PDF](#)

Zhibo Rao ; Yuchao Dai ; Zhelun Shen ; Renjie He [All Authors](#)

16
Cites in
Papers

763
Full
Text Views



Abstract

Document
Sections

- I. Introduction
- II. Related Work
- III. Methodology
- IV. Experiments

Abstract:

In stereo matching, various learning-based approaches have shown impressive performance in solving traditional difficulties on multiple datasets. While most progress is obtained on a specific dataset with a dataset-specific network design, the performance on the single dataset and cross dataset affected by training strategy is often ignored. In this article, we analyze the relationship between different training strategies and performance by retraining some representative state-of-the-art methods (e.g., geometry and context network (GC-Net), pyramid stereo matching network (PSM-Net), and guided aggregation network (GA-Net), etc.). According to our research, it is surprising that the performance of **IEEE Transactions on Pattern Analysis and Machine Intelligence** is improved by pre-training and data augmentation. Based on

How to find Research Topics

Attend Seminars and Conferences

Participate in academic conferences, seminars, and workshops. These events provide opportunities to learn about the latest research and network with experts who might offer inspiration for your own work



Attend research seminar or join a research group at AI Viet Nam



How to find Research Topics

Consider Current Events

Current events can also be a source of research topics. How do recent developments in the world relate to your field of interest, and what questions do they raise?



VIPriors Workshop | ICCV 2023

About Organizers Call for papers Challenge

Challenges

We present the “Visual Inductive Priors for Data-Efficient Computer Vision” challenges. We offer two challenges, where models are to be trained from scratch in a data-deficient setting:

1. Object detection
2. Instance segmentation

These tasks were chosen to encourage researchers of arbitrary background to participate: no giant GPU clusters are needed, nor will training for a long time yield much improvement over the baseline results.

OpenDriveLab

Welcome and Opening Remarks

E2EAD Workshop
CVPR 2023

CVPR 2023 | E2EAD Workshop

OpenDriveLab
12 videos • 9,224 views • Last updated on Jun 22, 2023

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1	CVPR23 E2EAD Welcome and Opening Remarks	1.1K views • 4 months ago	8:11
2	CVPR23 E2EAD Deva Ramanan, Invited Talk	1.4K views • 4 months ago	32:57
3	CVPR23 E2EAD Phil Duan, Invited Talk	10K views • 4 months ago	37:04
4	CVPR23 E2EAD 3D Occupancy Prediction Challenge	2.4K views • 4 months ago	54:33
5	CVPR23 E2EAD Patrick Langechuan Liu, Invited Talk	2.7K views • 4 months ago	39:37

Zalo Challenge /2023

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Challenges

ELEMENTARY MATHS SOLVING

Large Language Models are currently trending and hold significant potential for various applications in our daily lives. However, one of the challenges faced by these models and other AI systems is performing mathematical and reasoning...

ADVERTISING BANNER GENERATION

In the world of digital advertising, creating compelling product banners quickly is crucial. This competition challenges participants to develop an AI model capable of generating product advertising banners automatically, based on provide...

BACKGROUND MUSIC GENERATION

In the age of generative AI, the art of music composition is no longer confined solely to human composers. The AI music generation topic is designed to explore the creative potential of AI in the realm of background music creation. This challenge invites AI enthusiasts, data scientists, and musicians to harness the capabilities of AI to craft background music that enhances a myriad of contexts, from film and video games to advertising and online content.

How to find Research Topics

Collaborate with Others

Engage in discussions with peers and colleagues. They may have interesting ideas or be willing to collaborate on a research project.



How to find Research Topics

Narrow Down and Define

Stereoscopic Matching

Object Detection

After brainstorming and research, narrow your list of topics to a few specific areas of interest. Ensure that the topics are feasible and have enough available resources for research.

Object Tracking

Text-Image Retrieval

Generative AI

Graph Neural Network

...

...



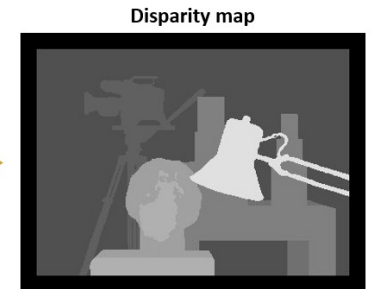
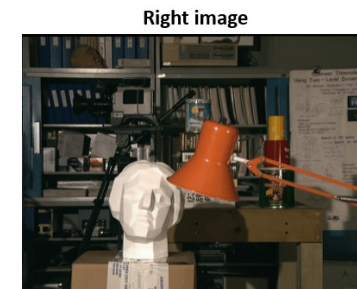
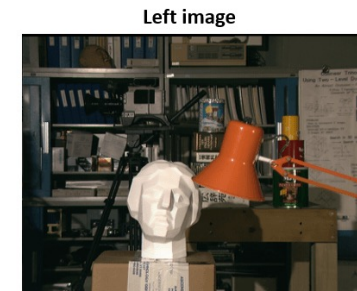
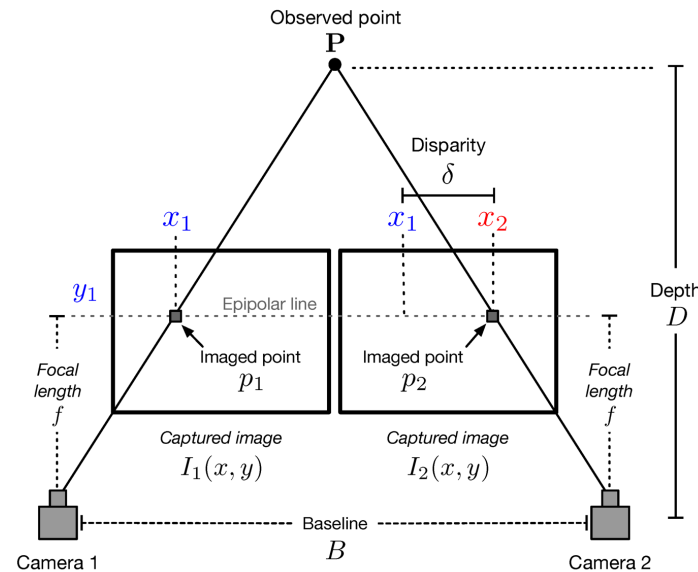
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Stereoscopic Matching



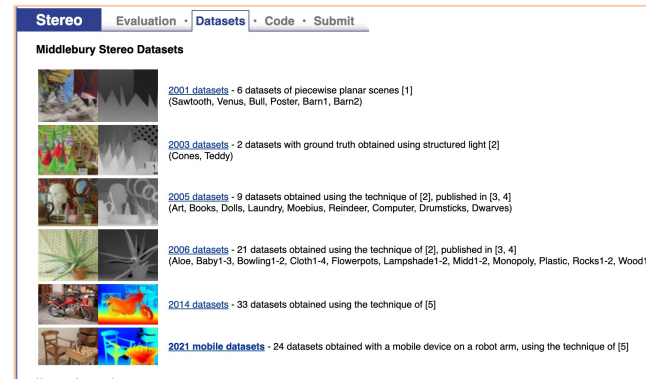
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Stereoscopic Matching



DrivingStereo: A Large-Scale Dataset for Stereo Matching in Autonomous Driving Scenarios

Publisher: IEEE

Cite This

PDF

Guorun Yang ; Xiao Song ; Chaoqin Huang ; Zhidong Deng ; Jianping Shi ; Bolei Zhou [All Authors](#)

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Cites in
Papers

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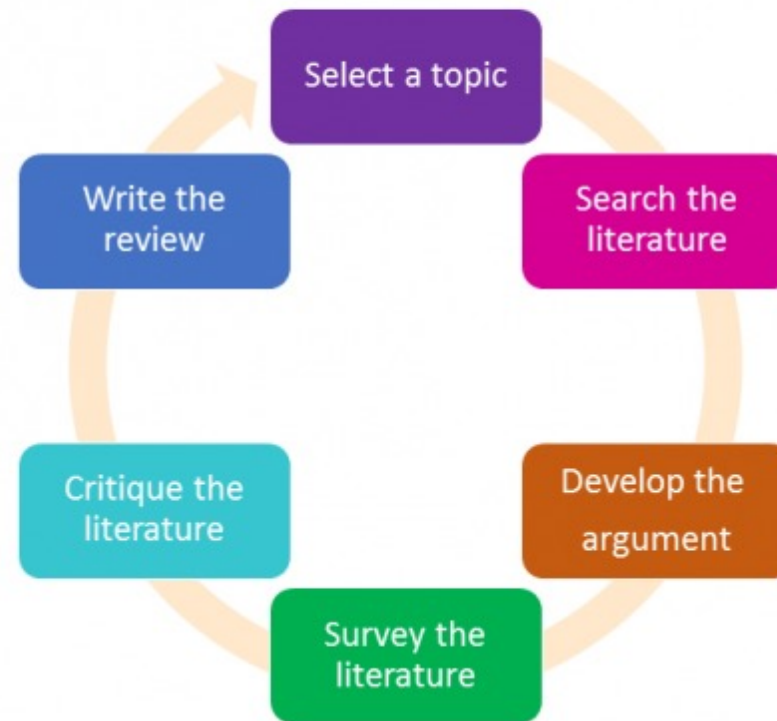


How to find Research Topics

Research the Literature

Before finalizing your topic, do a more in-depth literature review to make sure your chosen topic is not only interesting but also has a clear research gap.

A literature review is a critical analysis and synthesis of existing research and literature on a specific topic. The purpose of a literature review is to provide an overview of the current state of knowledge on a particular subject, identify gaps or areas where further research is needed, and establish the context for the new research being presented.



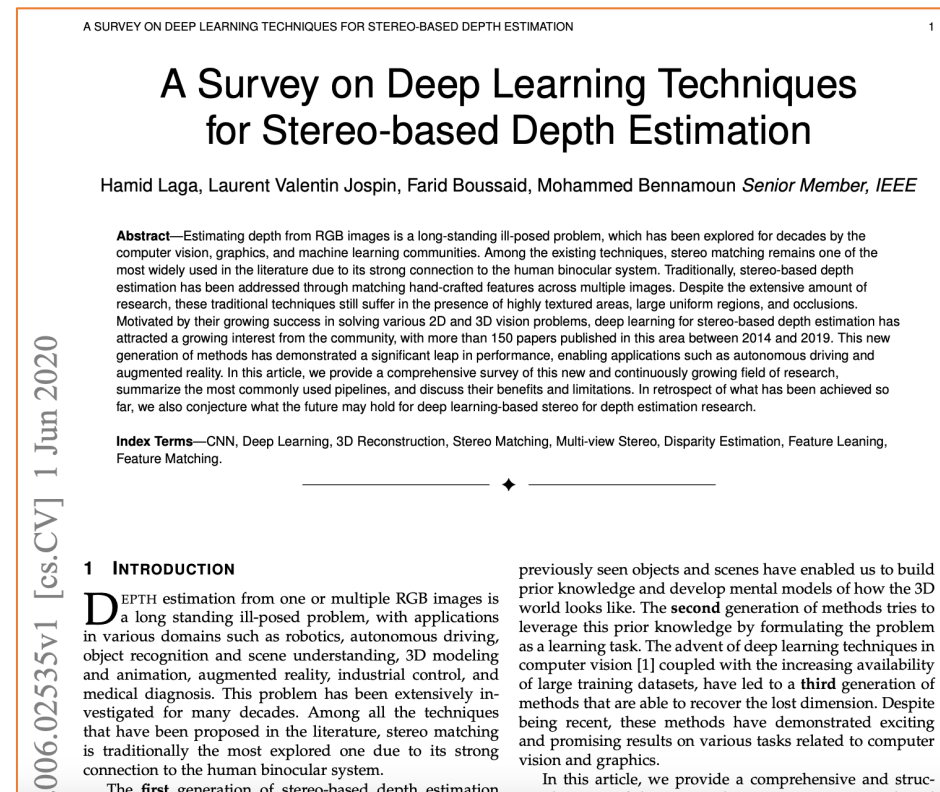
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TABLE 3: Taxonomy and comparison of 28 end-to-end deep learning-based disparity estimation techniques. "FCN": Fully-Connected Network, "SPN": Spatial Propagation Network. "LRCR": Left-Right Comparative Recurrent model, "MCUA": Multi-Level Context Ultra-Aggregation for Stereo Matching. "DLA": Deep layer aggregation [69], "VPP": Volumetric Pyramid Pooling. The performance is measured on KITTI2015 test dataset.

Method	Year	Feature computation		Type	Cost volume		Disparity	Refinement/post processing		Supervision	Performance D1-all/est D1-all/fg
		Architecture	Dimension		Construction	Regularization		Spatial/depth resolution	Completion/denoising		
FlowNetCorr [51]	2015	ConvNet	Single scale	3D	Correlation	2D ConvNet	Supervised	Up-convolutions	Ad-hoc, variational	Supervised	—
DispNetC [22]	2016	ConvNet	Single scale	3D	Correlation	2D ConvNet	—	—	—	Supervised	4.34 4.32
Zhong et al. [70]	2017	ConvNet with skip conn.	Single scale	4D	Interleaved feature concat.	3D Conv, encoder-decoder	Soft argmin	Self-improvement at runtime	—	Self-supervised	3.57 7.12
Kendall et al. [61]	2017	ConvNet with skip conn.	Single scale	4D	Feature concat.	3D Conv, encoder-decoder, hierarchical	Soft argmax	—	—	Supervised	2.87 6.16
Pang et al. [62]	2017	ConvNet	Single scale	3D	Correlation	2D ConvNet	—	Upsampling+ residual learning	—	Supervised	2.67 3.59
Knobler et al. [71]	2017	ConvNet	Single scale	3D	Correlation	Hybrid CNN-CRF	—	No post-processing	—	Supervised	—
Chang et al. [64]	2018	SPP	Multiscale	4D	Feature concat.	3D Conv, Stacked encoder-decoders	Soft argmin	Progressive refinement	—	Supervised	2.32 4.62
Khamis et al. [72]	2018	ResNet	Single scale	3D	L_2	3D ConvNet	Soft argmin	Hierarchical, Upsampling+residual learning	—	Supervised	4.83 7.45
Liang et al. [63]	2018	ConvNet	Multiscale	3D	Correlation	2D ConvNet	Encoder-decoder	Iterative upsampling+residual learning	—	Supervised	2.67 3.59
Yang et al. [68]	2018	Shallow ResNet	Single scale	3D	Correlation, Left features, segmentation mask	—	Regression with Encoder-decoder	—	—	Self-supervised	2.25 4.07
Zhang et al. [73]	2018	ConvNet	Single scale	3D	Hand-crafted	NA	Soft argmin	Upsampling+residual learning	—	Self-supervised	—
Jie et al. [74]	2018	Constant highway net	Single scale	3D	FCN	—	RNN-based LRCR	—	—	Supervised	3.03 5.42
Ilg et al. [75]	2018	ConvNet	Single scale	3D	Correlation	Encoder-decoder, joint disparity and occlusion	—	Cascade of encoder-decoders, residual learning	—	Supervised	2.19
Song et al. [76]	2018	SHallow ConvNet	Single scale	3D	Correlation Encoder	Edge-guided, Context Pyramid, Residual pyramid	—	—	—	Supervised	2.59 4.18
Yu et al. [77]	2018	ResNet	Single scale	3D	Feature concatenation Encoder-decoder	3D Conv + SGM	Soft argmin	—	—	Supervised	2.79 5.46
Tulyakov et al. [78]	2018	—	Single scale	4D	Compressed matching features	3D Conv	Multimodal - Sub-pixel MAP	—	—	Supervised	2.58 4.05
EMCUA et al. [65]	2019	SPP	Multiscale	4D	Feature concat.	3D Conv, MCUA scheme	Arg softmax	—	—	Supervised	2.09 4.27
Yang et al. [32]	2019	SPP	Multiscale	Pyramidal 4D	Feature difference	Conv3D blocks + Volume Pyramid Pooling	Arg softmax	Spatial & depth res. by cost volume upsampling	—	Supervised	2.14 3.85
Wu et al. [79]	2019	ResNet50, SPP	Multiscale	Pyramidal 4D	Feature concat.	Encoder-decoder + Feature fusion	3D Conv, soft argmin	—	—	Supervised	2.11 3.89
Yin et al. [80]	2019	DLA net	Multiscale	3D	Correlation	Density decoder	Outputs discrete matching distribution	—	—	Supervised	2.02 3.63
Chabra et al. [81]	2019	ConvNet + Vortex pooling [82]	Multiscale	3D	L_1	Dilated 3D ConvNet	Soft argmax	Upsampling+residual learning	—	Supervised	2.26 4.95
Duggal et al. [83]	2019	ResNet, SPP	Multiscale	3D, sparse	Correlation, Adaptive pruning with PatchMatch	Encoder-decoder	Soft argmax	Encoder	—	Supervised	2.35 3.43
Tonioni et al. [84]	2019	ConvNet	Multiscale	3D	Correlation	—	Encoder	Recursively upsampling + residual learning	—	Online self-adaptive	4.66
Yang et al. [32]	2019	ConvNet, SPP	Multiscale	Pyramid, 4D	Concatenation	Decoder, Residual blocks, VPP	Conv3D block	—	—	Supervised	2.14 3.85
Zhang et al. [85]	2019	Stacked hourglass	Single scale	4D	Concatenation	Semi-global aggregation layers, Local-guided aggregation layers	Soft argmax	—	—	Supervised	2.03 3.91
Guo et al. [86]	2019	SPP	Multiscale	Hybrid 3D-4D	Group-wise correlation	Stacked hourglass nets	Soft argmin	—	—	Supervised	2.11 3.93
Chen et al. [87]	2019	—	—	—	—	—	Single-modal weighted avg	—	—	Supervised	2.14 4.33
Wang et al. [88]	2019	ConvNet	Multiresolution maps	3D	L_1	Progressive refinement (3D Conv)	Soft argmin	Upsampling, residual learning	Spatial propagation network	Supervised	—

How to find Research Topics

Research the Literature

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Interestingly, Table 9 and Fig. 10 show that most of the methods achieve similar reconstruction accuracies. The only exception is iResNet [63] trained on Kitti2015 and on ROB [152], which had more than 90%, respectively 55%, of pixels with an error that is larger than 5 pixels. In all methods, less than 5% of the pixels had an error that is less than 2 pixels. This suggests that achieving sub-pixel accuracy remains an important challenge for future research.

Note that SegStereo [68], which is self-supervised, achieves a similar or better performance than many of the supervised methods. Also, the unsupervised self-adaptation method of Tonioni *et al.* [129], which takes the baseline DispNet-Corr1D network [22] trained on KITTI 2012 and adapts it to KITTI2015 and Middlebury 2014, achieves one of the best performances on the foreground regions.

In terms of the visual quality of the estimated disparities, see Fig. 11, we observe that most of the methods were able to recover the overall shape of trees but fail to reconstruct the details especially the leaves. The reconstruction errors are high in flat areas and around object boundaries. Also, highly reflective materials and poor lighting conditions remain a big challenge to these methods as shown in Fig. 11-(b). The supplementary material provides more results on the four stereo pairs of Fig. 9.

9 FUTURE RESEARCH DIRECTIONS

Deep learning methods for stereo-based depth estimation have achieved promising results. The topic, however, is still in its infancy and further developments are yet to be expected. In this section, we present some of the current issues and highlight directions for future research.

(1) *Camera parameters.* Most of the stereo-based techniques surveyed in this article require rectified images. Multi-view stereo techniques use Plane-Sweep Volumes or back-projected images/features. Both image rectification and

(2) *Lighting conditions and complex material properties.* Poor lighting conditions and complex materials properties remain a challenge to most of the current methods, see for example Fig. 11-(b). Combining object recognition, high-level scene understanding, and low-level feature learning can be one promising avenue to address these issues.

(3) *Spatial and depth resolution.* Most of the current techniques do not handle high resolution input images and generally produce depth maps of low spatial and depth resolution. Depth resolution is particularly limited, making the methods unable to reconstruct thin structures, *e.g.*, vegetation and hair, and structures located at a far distance from the camera. Although refinement modules can improve the resolution of the estimated depth maps, the gain is still small compared to the resolution of the input images. This has recently been addressed using hierarchical techniques, which allow on-demand reports of disparity by capping the resolution of the intermediate results [32]. In these methods, low resolution depth maps can be produced in realtime, and thus can be used on mobile platforms, while high resolution maps would require more computation time. Producing, in realtime, accurate maps of high spatial and depth resolutions remains a challenge for future research.

(4) *Realtime processing.* Most deep learning methods for disparity estimation use 3D and 4D cost volumes, which are processed and regularized using 2D and 3D convolutions. They are expensive in terms of memory requirements and processing time. Developing lightweight, and subsequently fast, end-to-end deep networks remains a challenging avenue for future research.

(5) *Disparity range.* Existing techniques uniformly discretize the disparity range. This results in multiple issues. In particular, although the reconstruction error can be small in the disparity space, it can result in an error of meters in the

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Rethinking Training Strategy in Stereo Matching

Publisher: IEEE

[Cite This](#)

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Zhibo Rao ; Yuchao Dai ; Zhelun Shen ; Renjie He [All Authors](#)

16

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Abstract

Document
Sections

- I. Introduction
- II. Related Work
- III. Methodology
- IV. Experiments

Abstract:

In stereo matching, various learning-based approaches have shown impressive performance in solving traditional difficulties on multiple datasets. While most progress is obtained on a specific dataset with a dataset-specific network design, the performance on the single dataset and cross dataset affected by training strategy is often ignored. In this article, we analyze the relationship between different training strategies and performance by retraining some representative state-of-the-art methods (e.g., geometry and context network (GC-Net), pyramid stereo matching network (PSM-Net), and guided aggregation network (GA-Net), etc.). According to our research, it is surprising that the performance of networks on single or cross datasets is significantly improved by pre-training and data augmentation without any particular structure acquirement. Based on

Outline

- Research Topics: Thing to Know
- How to Find a Good Research Topic
- How to Develop Research Questions
- Sample Research Topics in Computer Vision
- Assignment

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Stereo Matching – 3D Reconstruction – Depth Estimation

Camera parameters

Lighting conditions and complex material properties.

Spatial and depth resolution

Realtime processing.

Disparity range.

Training.

Automatically learning the network architecture, its activation functions, and its parameters from data



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Stereo Matching – 3D Reconstruction – Depth Estimation

~~Camera parameters~~

Lighting conditions and complex material properties.

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~~Training.~~

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How to find Research Topics

Finalize Research Questions

Based on the feedback, refine your research topic and question as needed. It's common for topics to evolve during this process.



Stereo Matching – 3D Reconstruction – Depth Estimation

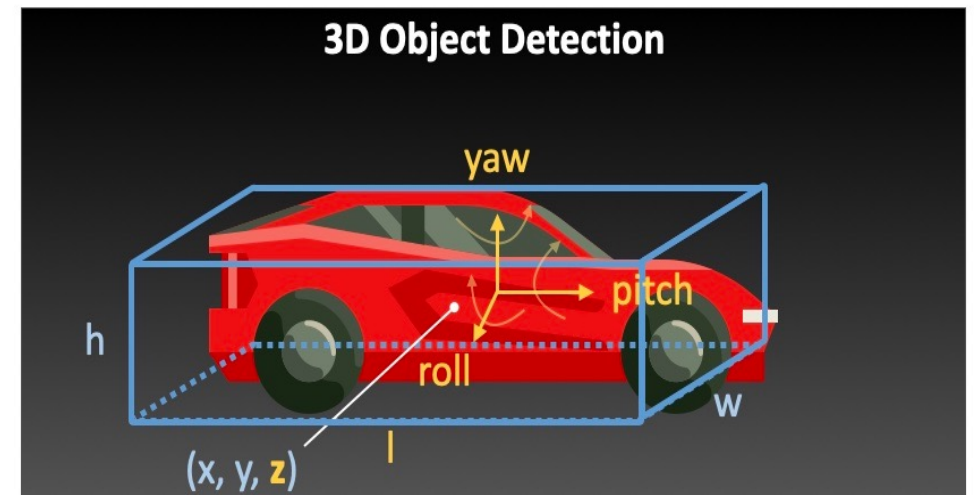
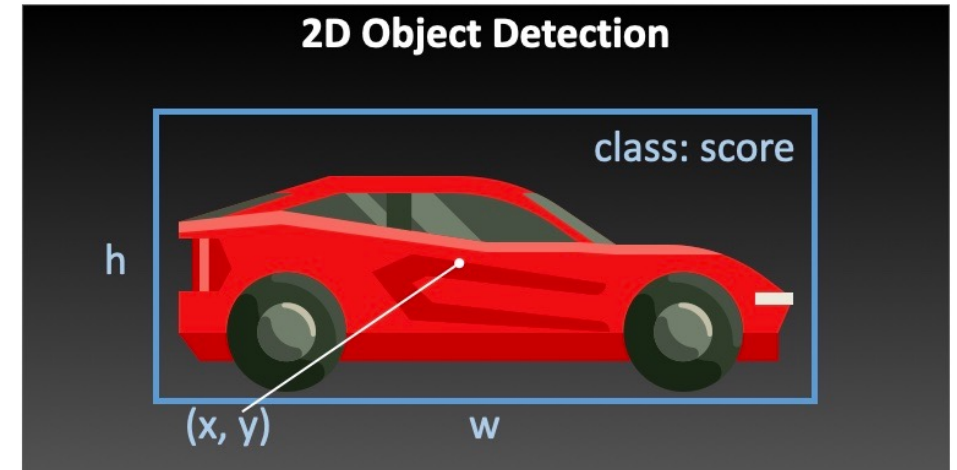
RQ1: How to improve the performance of existing stereo matching methods under various lighting conditions and complex material properties?

RQ2: How to obtain a real-time processing for current state-of-the-art stereo matching-based deep learning methods?

RQ3: How to design an automatically learning the network architecture, its activation functions, and its parameters from data ?

Remember that the best research topics are those that are not only of academic interest but also have real-world relevance and the potential to contribute to the existing body of knowledge. It's essential to choose a topic that you are genuinely passionate about and willing to invest time and effort into exploring.

Case study 2: 3D Object Detection



How to find Research Topics

Research the Literature

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IJCV manuscript No.
(will be inserted by the editor)

Multi-Modal 3D Object Detection in Autonomous Driving: A Survey

Yingjie Wang* · Qiuyu Mao* · Hanqi Zhu · Jiajun Deng · Yu Zhang ·
Jianmin Ji · Houqiang Li · Yanyong Zhang

Received: date / Accepted: date

Abstract The past decade has witnessed the rapid development of autonomous driving systems. However, it remains a daunting task to achieve full autonomy, especially when it comes to understanding the ever-changing, complex driving scenes. To alleviate the difficulty of perception, self-driving vehicles are usually equipped with a suite of sensors (*e.g.*, cameras, LiDARs), hoping to capture the scenes with overlapping perspectives to minimize blind spots. Fusing these data streams and exploiting their complementary properties is thus rapidly becoming the current trend.

Nonetheless, combining data that are captured by different sensors with drastically different ranging/imagining mechanisms is not a trivial task; instead, many factors need to be considered and optimized. If not careful, data from one sensor may act as noises to data from another sensor, with even poorer results by fusing them. Thus far, there has been no in-depth guidelines to designing the multi-modal fusion based 3D perception algorithms. To fill in the void and motivate further investigation, this survey conducts a thorough study of tens of recent deep learning based multi-modal 3D detection networks (with a special emphasis on LiDAR-camera fusion), focusing on their fusion stage (*i.e.*, when to fuse), fusion inputs (*i.e.*, what to fuse), and fusion granularity (*i.e.*, how to fuse). These important design choices play a critical role in determining the performance of the fusion algorithm.

*equal contribution

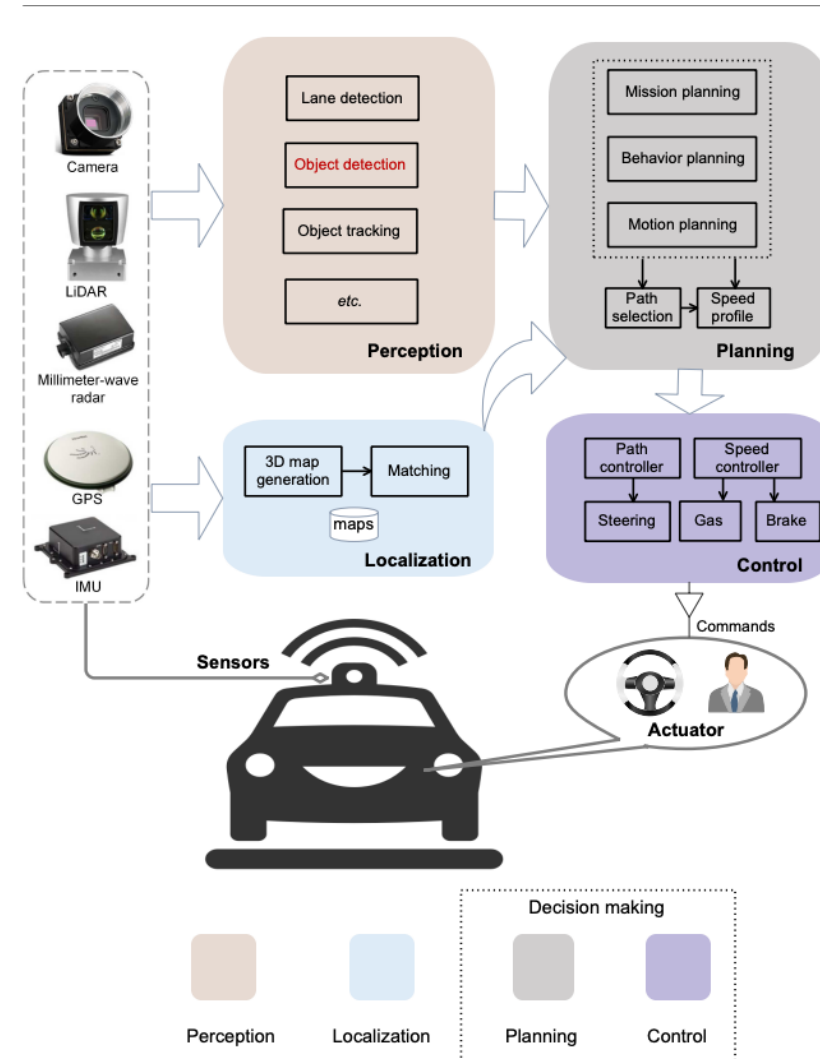
Yingjie Wang
University of Science and Technology of China
E-mail: yingjiawang@mail.ustc.edu.cn

12735v3 [cs.CV] 7 Mar 2023

How to find Research Topics

Research the Literature

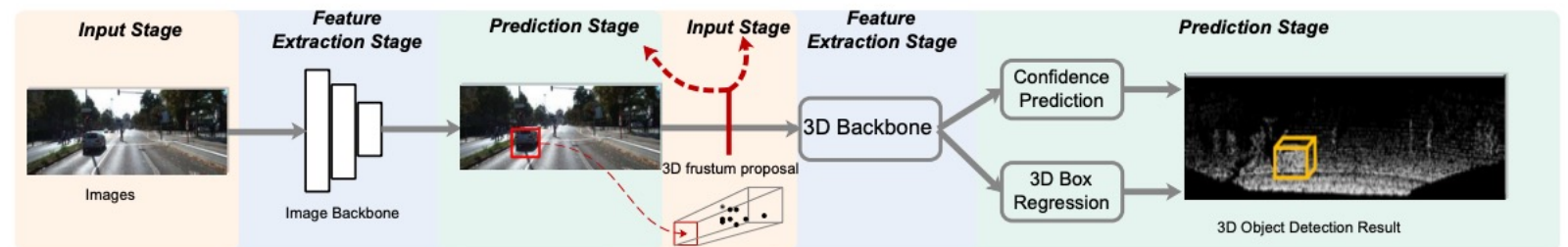
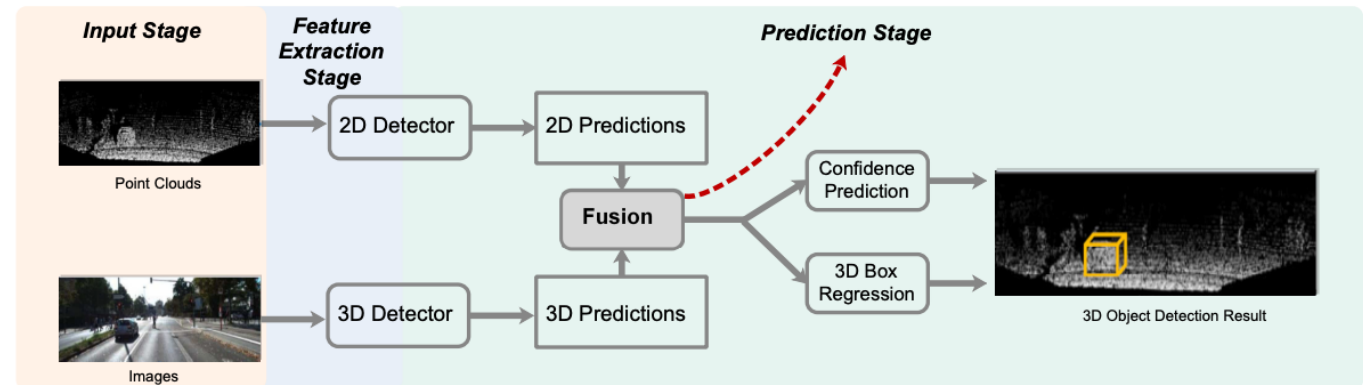
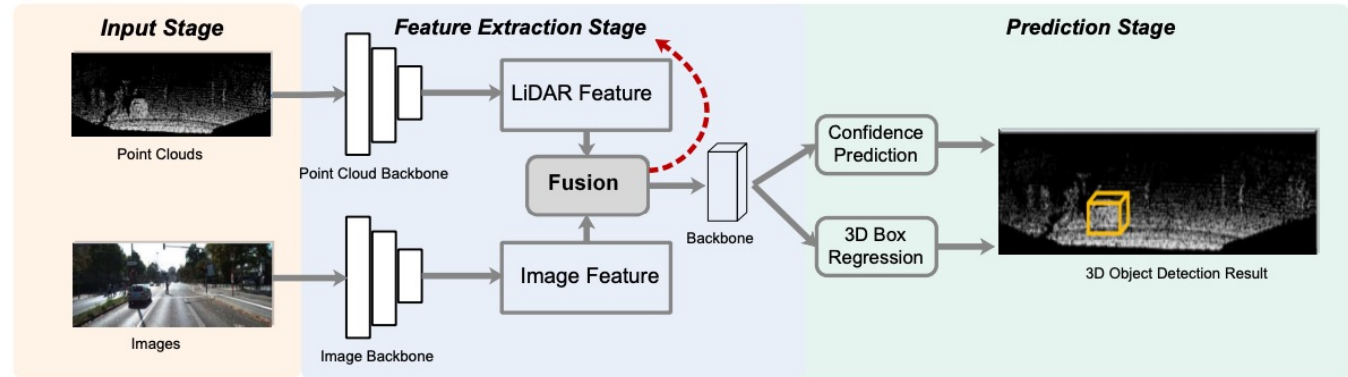
Before finalizing your topic, do a more in-depth literature review to make sure your chosen topic is not only interesting but also has a clear research gap.



How to find Research Topics

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Before finalizing your topic, do a more in-depth literature review to make sure your chosen topic is not only interesting but also has a clear research gap.



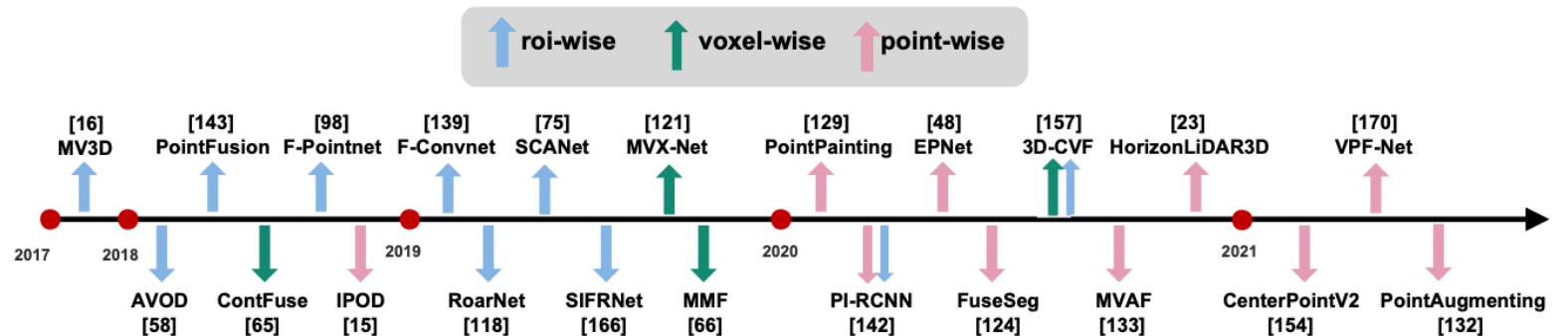
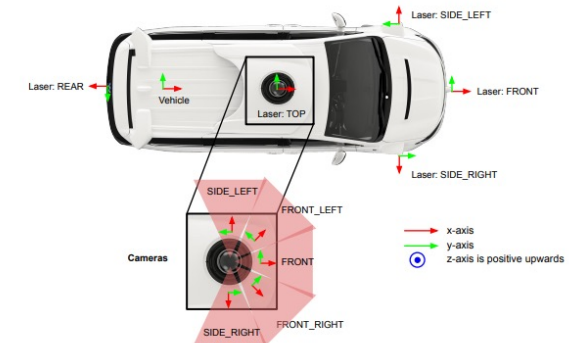
How to find Research Topics

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Before finalizing your topic, do a more in-depth literature review to make sure your chosen topic is not only interesting but also has a clear research gap.



	stage	PC-Input	RGB-Input	gran	HW	lat	DS	mAP
MV3D [19]	early	view	feature map	RoI	Titan X	0.36s	KITTI	63.63%
AVOD [78]				RoI	Titan XP	0.08s	KITTI	71.76%
Confuse [85]				voxel	GTX1080	0.06s	KITTI	68.78%
SCANet [98]				RoI	GTX1080	0.09s	KITTI	66.30%
FuseSeg [151]				point	-	-	KITTI	-
MVX-Net [148]		voxel	feature map	voxel	GTX1080	-	KITTI	72.70%
3D-CVF [187]				voxel & RoI	-	0.06s	KITTI	80.45%
VPF-Net [200]				point	2080Ti	0.06s	KITTI	83.21%
PointAugmenting [159]				point	-	-	nuScenes	66.80%
PointFusion [173]		point	feature map	RoI	GTX1080	1.3s	KITTI	63.00%
EPNet [65]				point	Titan XP	0.1s	KITTI	81.23%
PI-RCNN [172]				point & RoI	-	0.06s	KITTI	78.53%
PointPainting [156]	late	point	mask	point	GTX1080	0.4s	KITTI	75.80%
CenterPointV2 [185]				point	-	-	nuScenes	67.10%
HorizonLiDAR3D [32]				point	-	-	Waymo	78.49%
MMF [86]		view	feature map & pseudo LiDAR	point	GTX1080	0.08s	KITTI	77.43%
MVAF [160]		voxel & view	feature map	voxel	Titan X	0.06s	KITTI	78.71%
F-PointNet [123]	cascade	-	-	RoI	GTX1080	0.17s	KITTI	69.79%
IPOD [181]					-	0.1s	KITTI	72.57%
F-ConvNet [167]					-	0.1s	KITTI	75.50%
RoarNet [145]					Titan X	-	KITTI	73.04%
SIFRNet [196]					-	-	KITTI	-
CLOCs [112]	late	-	-	-	-	-	KITTI	82.25%



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5.1 Open Challenge I: Multi-Sensor Calibration

As shown in Fig. 25, multiple sensors mounted on the autonomous vehicle are from different sensor coordinates. Fusion based methods are required the alignment of these sensor data. Here, we use LiDAR-camera fusion as example to explain this challenge. Point clouds are a set of points indicating 3D coordinates of the objects. RGB images are matrices of pixels, with each pixel's coordinate represented as (x, y) , where x, y is the pixel's row and column index, respectively. To build the map from 3D LiDAR coordinates to the 2D image plane, we must perform calibration between the two.

Traditional calibration methods use a calibration target to derive the intrinsic and extrinsic camera parameters. This cumbersome process requires lots of manual efforts. A common practice is to develop a target-less, automatic calibration method that can continuously calibrate the LiDAR sensor and camera on the fly. Target-less calibration is currently an active topic of research in this field [82, 111, 139]. These methods automatically calibrate among multiple sensors without human experts. However, inevitable bumps and jitters when driving AVs lead to the variation of the extrinsic parameters for the well calibrated LiDAR-camera system. Much worse, the error will gradually accumulate if not corrected in time, which may eventually affect the perception results. A possible solution to prevent this problem is to integrate the LiDAR and camera in a suite [66, 150], preventing their relative displacement to the greatest extent.

5.2 Open Challenge II: Information Loss during Fusion

When fusing data from multiple modalities, a certain amount of information will be lost inevitably due to projection, quantization, feature blurring, *etc.* When devising a multi-modal fusion network, we need to pay attention to the stage, input, and granularity of the fusion operation, in order to minimize the information loss.

5.3 Open Challenge III: Efficient Multi-Modal Data Augmentation

Due to the limited number of objects in the dataset, data augmentation is usually adopted to ensure efficient learning and avoid overfitting. Existing data augmentation techniques for each single data stream can be applied to deep fusion methods, such as object cut-and-paste, random flipping, scaling, rotation, and so

5.4 Open Challenge IV: Low-Cost Multi-Modal 3D Object Detection

Monocular or stereo cameras are the most common low-cost sensors that can meet the requirements of mass production. However, without accurate 3D geometry information, relying on cameras alone cannot yield 3D detection results comparable to LiDAR-based methods. In fact, the state-of-the-art monocular method DD3D [113] achieves only 16.87% mAP on the KITTI 3D object detection leader board; the best stereo method LIGA-Stereo [52] can achieve 64.66% mAP. Nevertheless, the best LiDAR-only method BtcDet [174] has obtained 82.86% mAP.

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3D Object Detection

Multi-Sensor Calibration

Information Loss during Fusion

Efficient Multi-Modal Data Augmentation

Low-Cost Multi-Modal 3D Object Detection

Shortage of Large Datasets



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3D Object Detection

~~Multi-Sensor Calibration~~

Information Loss during Fusion

Efficient Multi-Modal Data Augmentation

Low-Cost Multi-Modal 3D Object Detection

~~Shortage of Large Datasets~~

How to find Research Topics

Finalize Research Questions

Based on the feedback, refine your research topic and question as needed. It's common for topics to evolve during this process.



3D Object Detection

RQ1: How to reduce Information Loss during Fusion stage?

RQ2: How to select optimal fusion inputs?

RQ3: How to design a robust method for Multi-Modal Data Augmentation?

RQ3: How to a robust method for Low-Cost Multi-Modal 3D Object Detection?

Case study 3: Text-Image Retrieval

Image retrieval system

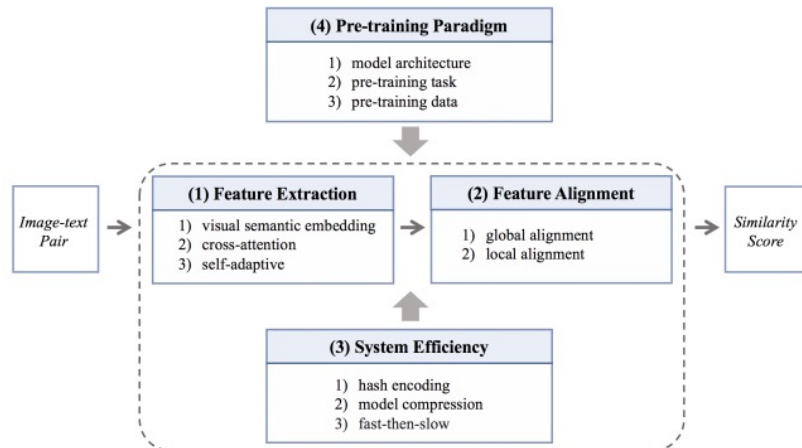
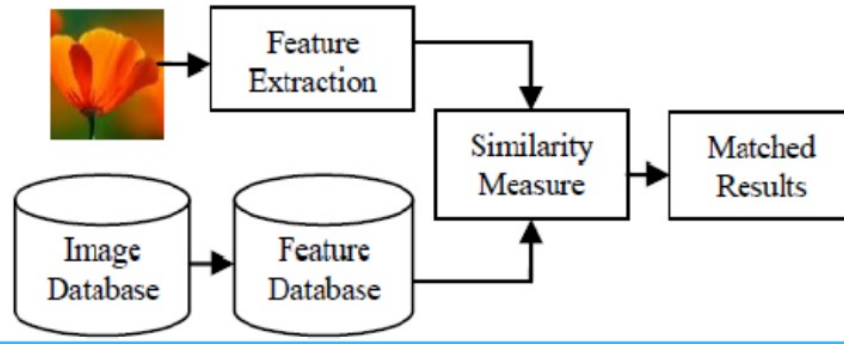


Image-text Retrieval: A Survey on Recent Research and Development

Min Cao¹, Shiping Li¹, Juntao Li¹, Liqiang Nie² and Min Zhang^{1,3}

¹Soochow University, ²Shandong University, ³Harbin Institute of Technology, Shenzhen
{mcao,ljt,minzhang}@suda.edu.cn, spli@stu.suda.edu.cn, nieliqiang@gmail.com

Abstract

In the past few years, cross-modal image-text retrieval (ITR) has experienced increased interest in the research community due to its excellent research value and broad real-world application. It is designed for the scenarios where the queries are from one modality and the retrieval galleries from another modality. This paper presents a comprehensive and up-to-date survey on the ITR approaches from four perspectives. By dissecting an ITR system into two processes: feature extraction and feature alignment, we summarize the recent advance of the ITR approaches from these two perspectives. On top of this, the efficiency-focused study on the ITR system is introduced as the third perspective. To keep pace with the times, we also provide a pioneering overview of the cross-modal pre-training ITR approaches as the fourth perspective. Finally, we outline the common benchmark datasets and evaluation metric for ITR, and conduct the accuracy comparison among the representative ITR approaches. Some critical yet less studied issues are discussed at the end of the paper.

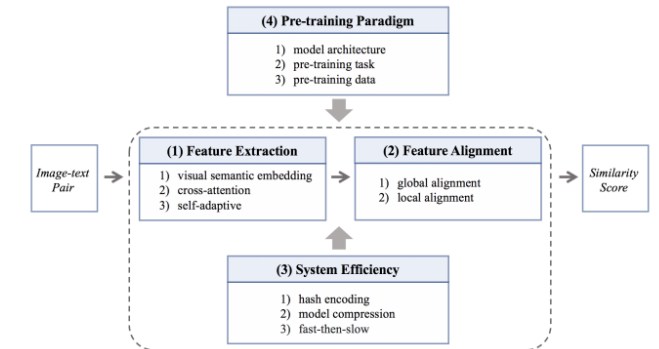


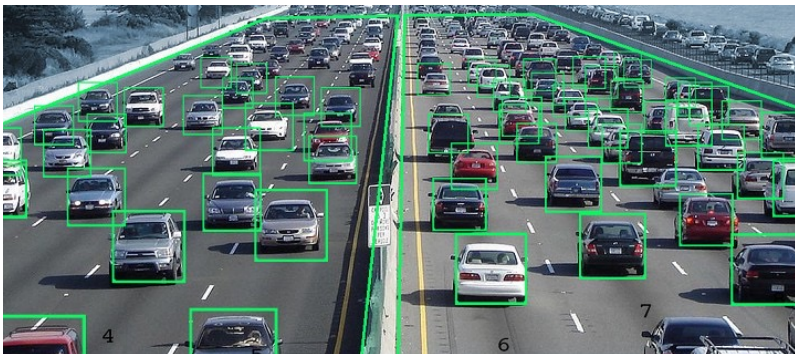
Figure 1: Illustration of the classification skeleton of ITR approaches from four perspectives.

less in-depth ITR survey [Uppal *et al.*, 2022; Baltrušaitis *et al.*, 2018]; 2) the pre-training paradigm is largely untapped in existing surveys [Chen *et al.*, 2020b] and is indeed the mainstream nowadays. In light of these, we conduct a comprehensive and up-to-date survey on the ITR task in this paper, especially with an investigation on the pre-training paradigm.

An ITR system generally consists of the feature extraction process with the image/text processing branches and the fea-

3.14713v3 [cs.IR] 18 Nov 2022

Case study 3: 2D Vehicle Detection



A Review of Vehicle Detection Techniques for Intelligent Vehicles

Publisher: IEEE

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Zhangu Wang ; Jun Zhan ; Chunguang Duan ; Xin Guan ; Pingping Lu ; Kai Yang [All Authors](#)

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Abstract

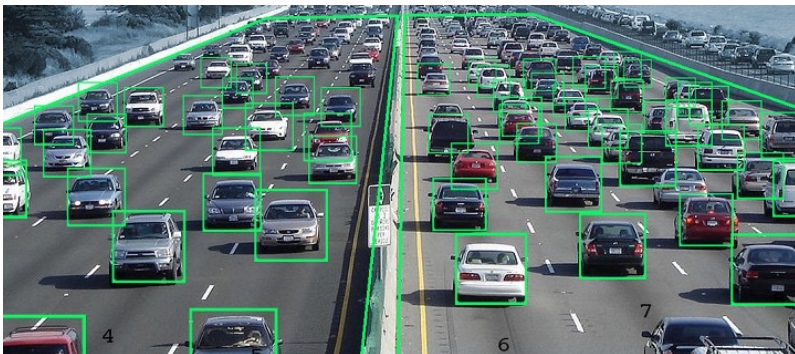
Document Sections

- I. Introduction
- II. Vehicle Detection Method
Based on Machine Vision
- III. Vehicle Detection Method
Based on Millimeter-Wave
Radar
- IV. Vehicle Detection Method
Based on LIDAR

Abstract:

Robust and efficient vehicle detection is an important task of environment perception of intelligent vehicles, which directly affects the behavior decision-making and motion planning of intelligent vehicles. Due to the rapid development of sensor and computer technology, the algorithm and technology of vehicle detection have been updated rapidly. But, there are few reviews on vehicle detection of intelligent vehicles, especially covering all kinds of sensors and algorithms in recent years. This article presents a comprehensive review of vehicle detection approaches and their applications in intelligent vehicle systems to analyze the development of vehicle detection, with a specific focus on sensor types and algorithm classification. First, more than 300 research contributions are summarized in this review, including all kinds of vehicle detection sensors (machine vision, millimeter-wave radar, lidar, and multisensor fusion), and the performance of the classic and latest algorithms was compared in detail. Then, the application scenarios of vehicle detection with different sensors and algorithms were analyzed according to their performance and applicability. Moreover, we also systematically summarized the methods of vehicle detection in adverse weather. Finally, the remaining challenges and future research trends were analyzed according to the development of intelligent vehicle sensors and algorithms.

Case study 3: 2D Vehicle Detection



The KITTI Vision Benchmark Suite

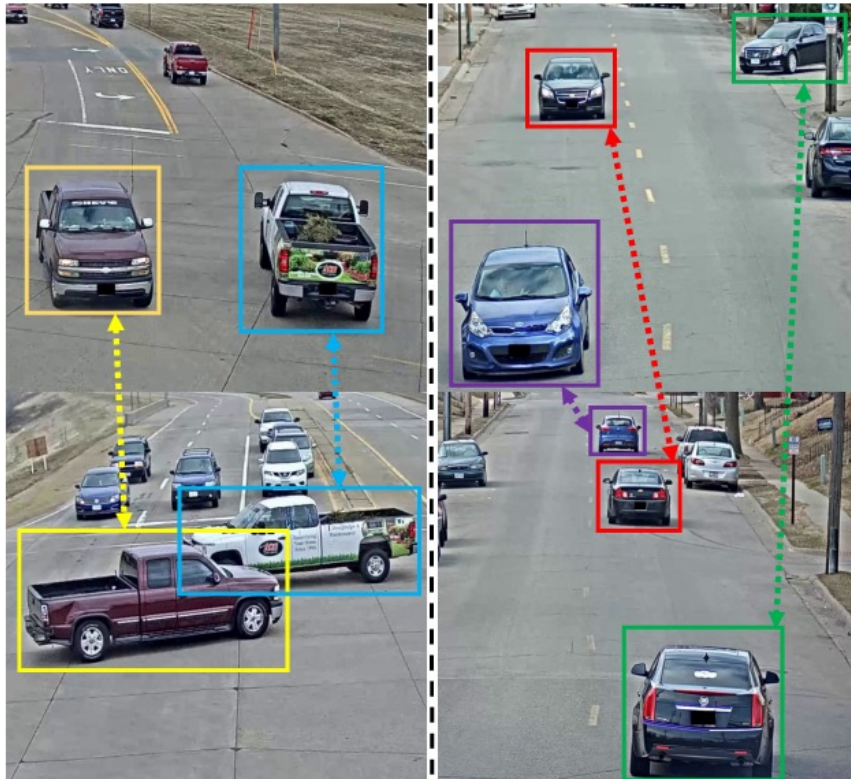
A project of Karlsruhe Institute of Technology and Toyota Technological Institute at Chicago



Car

	Method	Setting	Code	Moderate	Easy	Hard	Runtime	Environment	Compare
1	LVP(84.92)			97.84 %	98.70 %	93.07 %	0.04 s	1 core @ 2.5 Ghz (C/C++)	☐
2	HPC-Net			97.59 %	98.61 %	93.01 %	0.18 s	1 core @ 2.5 Ghz (C/C++)	☐
3	UDeerPEP		code	97.43 %	98.42 %	94.94 %	0.1 s	1 core @ 2.5 Ghz (C/C++)	☐
Z. Dong, H. Ji, X. Huang, W. Zhang, X. Zhan and J. Chen: PeP: a Point enhanced Painting method for unified point cloud tasks . 2023.									
4	VirConv-S		code	97.27 %	98.00 %	94.53 %	0.09 s	1 core @ 2.5 Ghz (C/C++)	☐
H. Wu, C. Wen, S. Shi and C. Wang: Virtual Sparse Convolution for Multimodal 3D Object Detection . CVPR 2023.									
5	GraR-Vol		code	96.38 %	96.81 %	91.20 %	0.07 s	1 core @ 2.5 Ghz (Python + C/C++)	☐
H. Yang, Z. Liu, X. Wu, W. Wang, W. Qian, X. He and D. Cai: Graph R-CNN: Towards Accurate 3D Object Detection with Semantic-Decorated Local Graph . ECCV 2022.									
6	VirConv-T		code	96.38 %	98.93 %	93.56 %	0.09 s	1 core @ 2.5 Ghz (C/C++)	☐
H. Wu, C. Wen, S. Shi and C. Wang: Virtual Sparse Convolution for Multimodal 3D Object Detection . CVPR 2023.									

Case study 4: Object Tracking

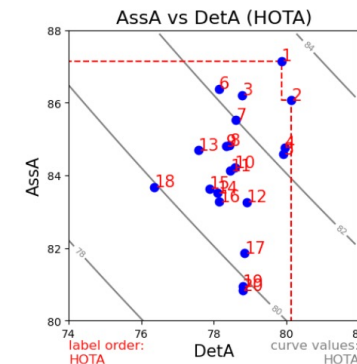


The KITTI Vision Benchmark Suite

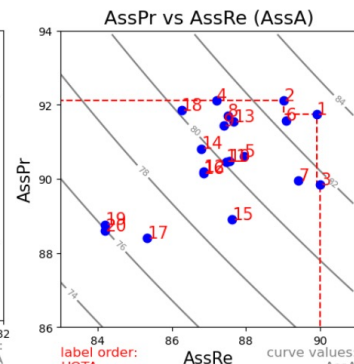
A project of Karlsruhe Institute of Technology and Toyota Technological Institute at Chicago



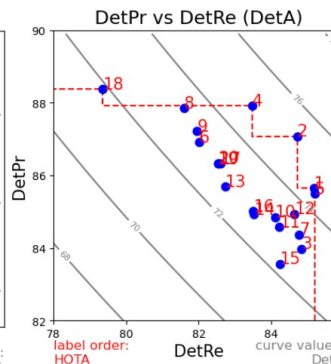
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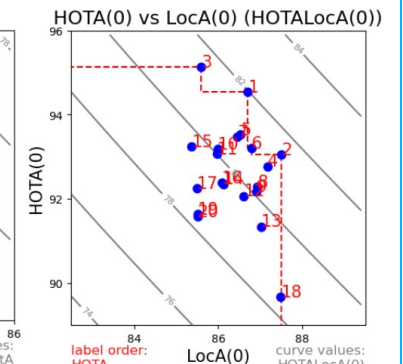
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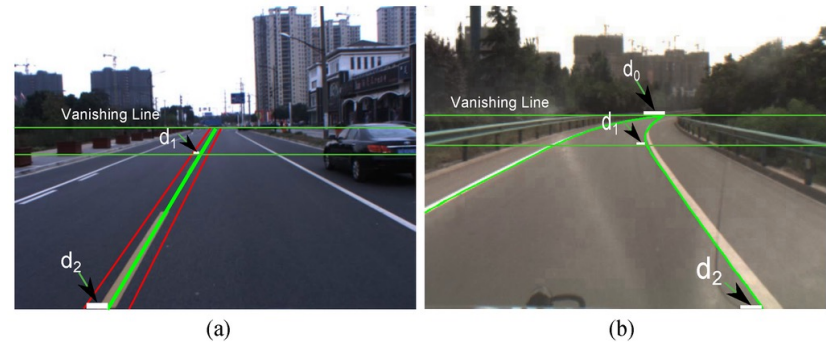


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	Method	Setting	Code	HOTA	DetA	AssA	DetRe	DetPr	AssRe	AssPr	LocA	MOTA	Compare
1	HRI-SFMOT			83.04 %	79.87 %	87.15 %	85.17 %	85.65 %	89.91 %	91.74 %	87.99 %	92.62 %	☐
2	BiTrack			82.69 %	80.13 %	86.07 %	84.71 %	87.08 %	89.03 %	92.11 %	88.65 %	91.65 %	☐
3	IMOU_ALG			82.08 %	78.78 %	86.21 %	84.83 %	83.98 %	90.00 %	89.84 %	87.14 %	92.75 %	☐
4	CrossTracker			82.00 %	79.96 %	84.75 %	83.48 %	87.92 %	87.20 %	92.12 %	88.39 %	91.97 %	☐
5	FusionTrack			81.90 %	79.92 %	84.59 %	85.19 %	85.50 %	87.95 %	90.61 %	87.87 %	92.54 %	☐
6	VirConvTrack		code	81.87 %	78.14 %	86.39 %	82.00 %	86.92 %	89.08 %	91.58 %	88.04 %	90.24 %	☐

H. Wu, C. Wang, S. Ermon, C. Wang, Virtual Energy Generation for Multimodal 3D Object Detection, CVPR 2022

Case study 5: Road Estimation



The KITTI Vision Benchmark Suite
A project of Karlsruhe Institute of Technology and Toyota Technological Institute at Chicago



	Method	Setting	Code	MaxF	AP	PRE	REC	FPR	FNR	Runtime	Environment	Compare
1	RD-RoTr			97.28 %	92.66 %	97.37 %	97.19 %	1.20 %	2.81 %	0.02 s	1 GPU @ 2.5 Ghz (Python)	☐
2	HF²Net			97.25 %	93.52 %	97.48 %	97.03 %	1.14 %	2.97 %	0.03 s	GPU @ 2.5 Ghz (Python)	☐
3	PLARD		code	97.05 %	93.53 %	97.18 %	96.92 %	1.28 %	3.08 %	0.16 s	GPU @ 2.5 Ghz (Python)	☐
Z. Chen, J. Zhang and D. Tao: Progressive LiDAR adaptation for road detection . IEEE/CAA Journal of Automatica Sinica 2019.												
4	RoadFormer			97.02 %	93.34 %	96.84 %	97.20 %	1.45 %	2.80 %	0.07 s	GPU @ 2.5 Ghz (Python)	☐
J. Li, Y. Zhang, P. Yun, G. Zhou, Q. Chen and R. Fan: RoadFormer: Duplex Transformer for RGB-Normal Semantic Road Scene Parsing . 2023.												
5	SNE-RoadSeg+			96.95 %	93.60 %	96.99 %	96.90 %	1.37 %	3.10 %	0.08 s	GPU @ 2.5 Ghz (Python)	☐
H. Wang, R. Fan, P. Cai and M. Liu: SNE-RoadSeg+: Rethinking depth-normal translation and deep supervision for freespace detection . 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2021.												
6	PLB-RD			96.87 %	93.71 %	97.35 %	96.40 %	1.20 %	3.60 %	0.46 s	GPU @ 2.5 Ghz (Python)	☐
L. Sun, H. Zhang and W. Yin: Pseudo-LiDAR-Based Road Detection . IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT) 2022.												
7	CLCFNet-SD			96.57 %	90.71 %	96.65 %	96.49 %	1.53 %	3.51 %	0.05 s	GPU @ 2.5 Ghz (Python)	☐
8	USNet		code	96.46 %	92.78 %	96.32 %	96.60 %	1.68 %	3.40 %	0.02 s	GPU @ 1.5 Ghz (Python)	☐
Y. Chang, F. Xue, F. Sheng, W. Liang and A. Ming: Fast Road Segmentation via Uncertainty-aware Symmetric Network . IEEE International Conference on Robotics and Automation (ICRA) 2022.												
9	DFM-RTFNet			96.46 %	93.66 %	96.58 %	96.33 %	1.55 %	3.67 %	0.08 s	GPU @ 2.5 Ghz (Python)	☐
H. Wang, R. Fan, Y. Sun and M. Liu: Dynamic fusion module evolves drivable area and road anomaly detection: A benchmark and algorithms . IEEE Transactions on Cybernetics 2021.												
10	SNE-RoadSeg		code	96.42 %	93.67 %	96.59 %	96.26 %	1.55 %	3.74 %	0.18 s	GPU @ 2.5 Ghz (Python)	☐
R. Fan, H. Wang, P. Cai and M. Liu: SNE-RoadSeg: Incorporating Surface Normal Information into Semantic Segmentation for Accurate Freespace Detection . ECCV 2020.												

Verifying the feasibility of research topics

Literature Review

Expert Consultation

Preliminary Research

Available Resources

Technological and Methodological Viability

Time Frame

Ethical and Legal Considerations

Collaboration Opportunities

Risk Assessment

Feasibility Study:

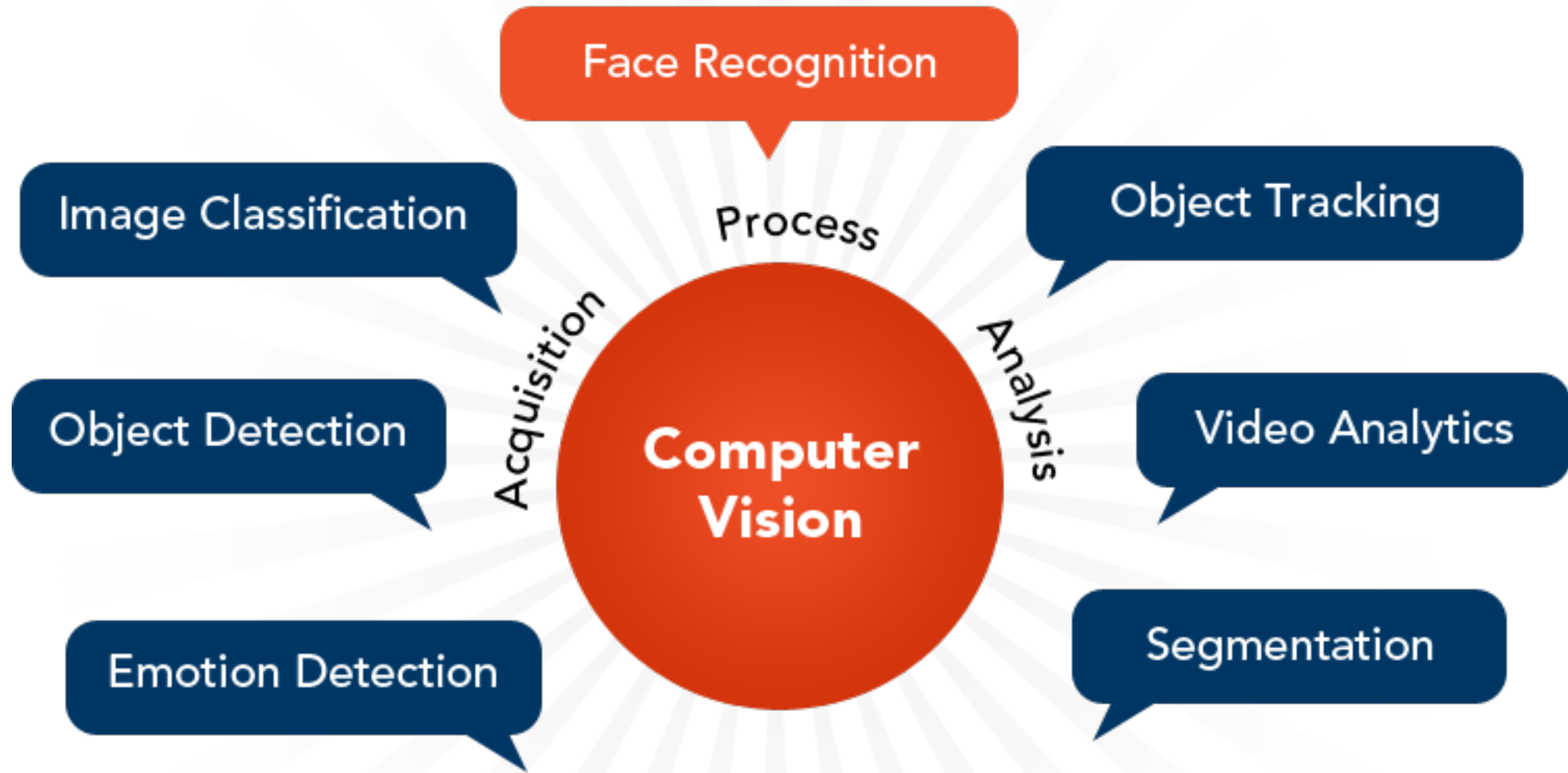
Feedback and Peer Review:

Iterative Process: prepared to adapt and modify your research directions

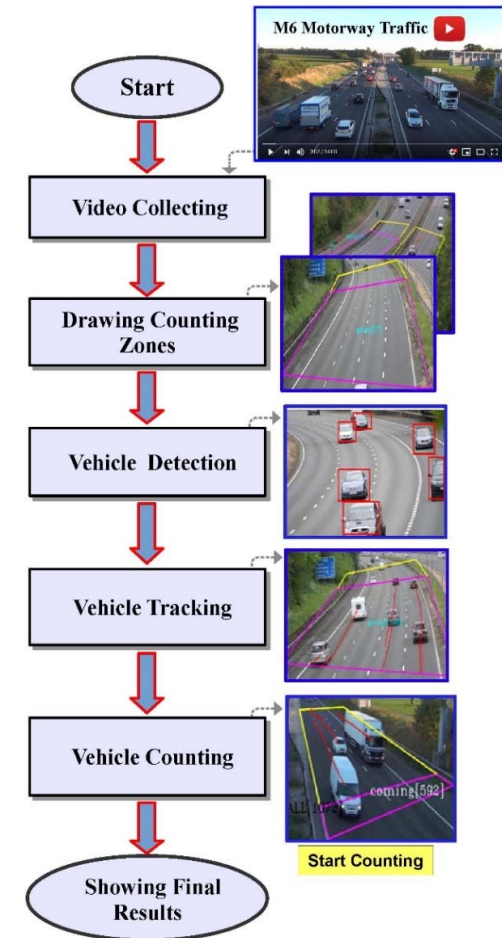
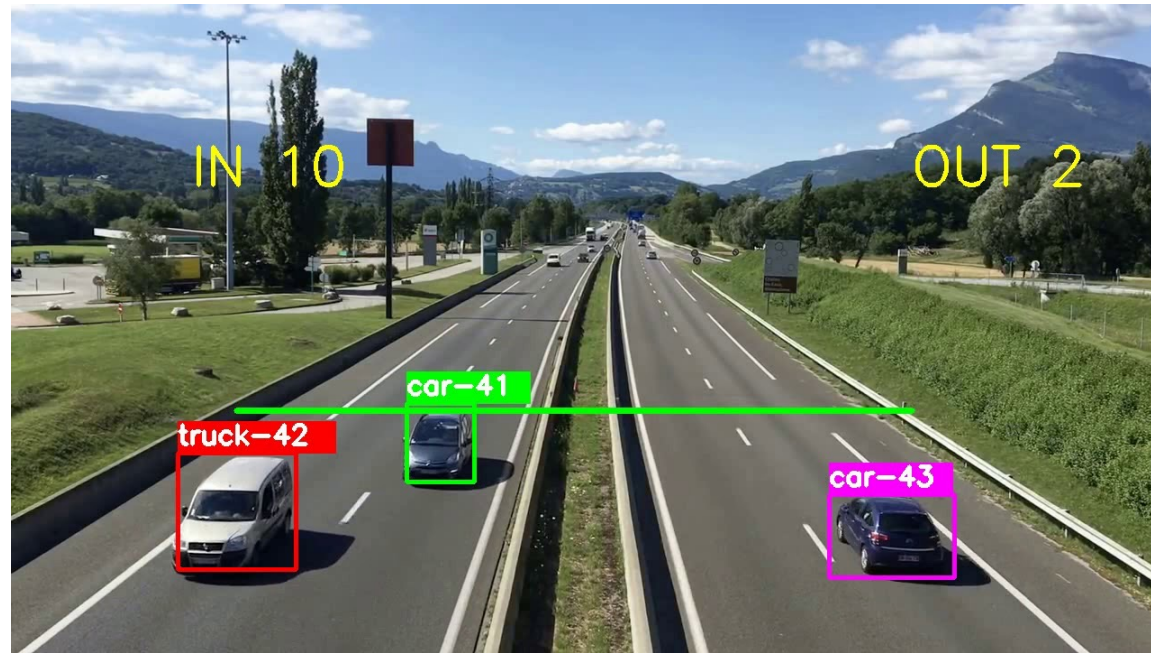
Outline

- **Research Topics: Thing to Know**
- **How to Find a Good Research Topic**
- **How to Develop Research Questions**
- **Sample Research Topics in Computer Vision**
- **Assignment**

Research Topics in Computer Vision



Vehicle Counting

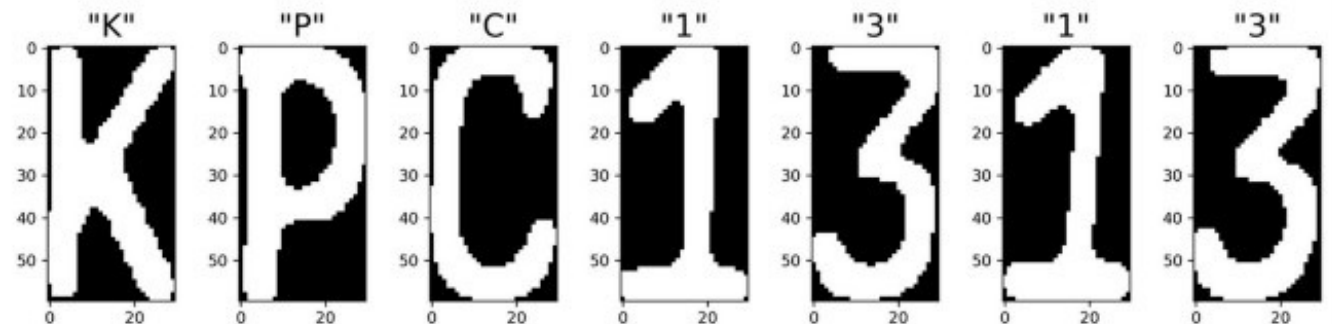


Car License Plate Detection

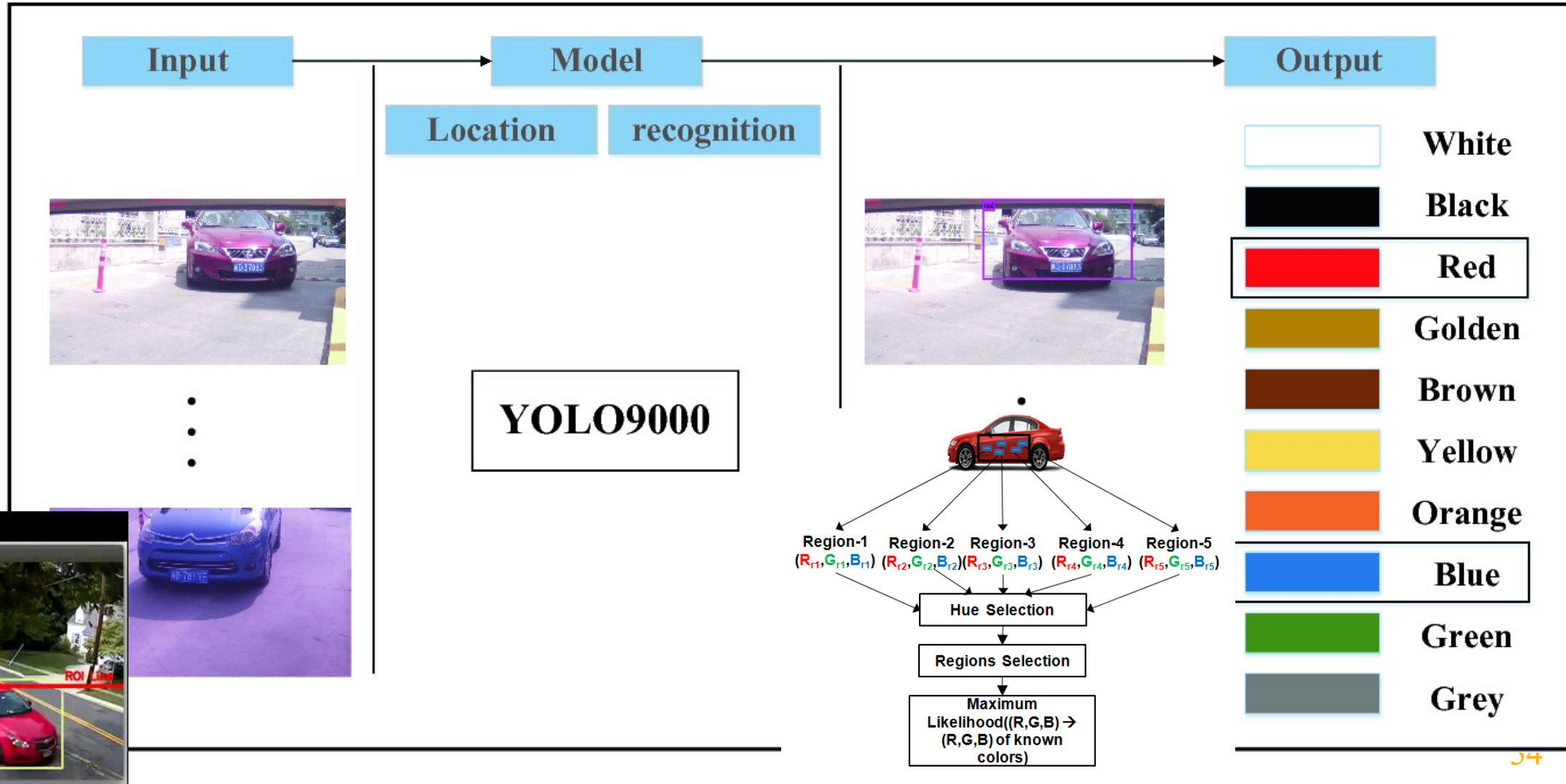
(Chiều cao ký tự trên biển số >16pixels)



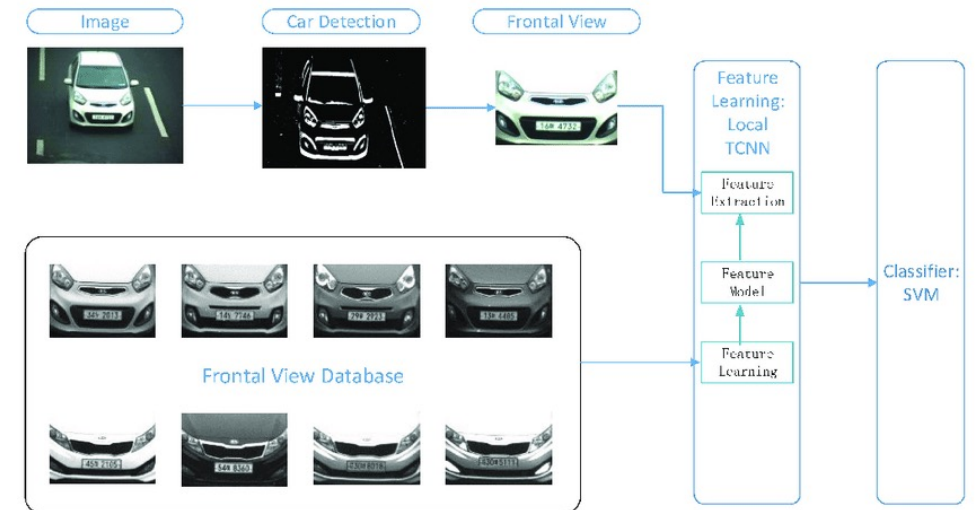
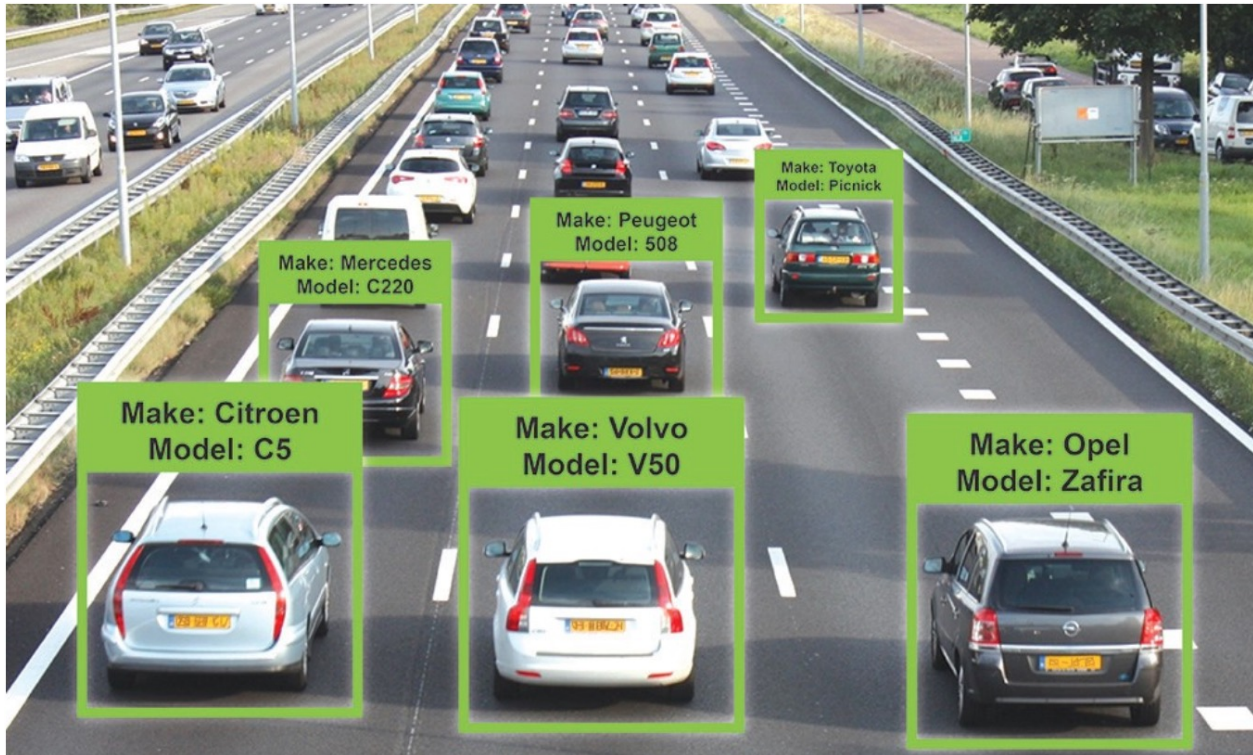
KPC1313



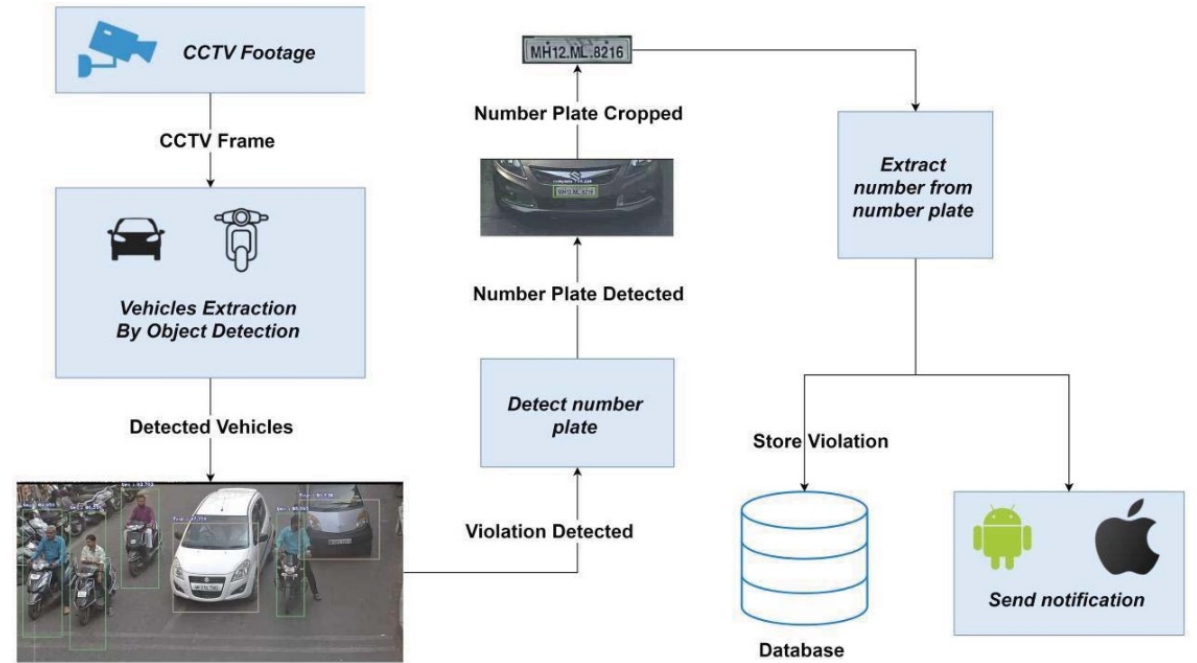
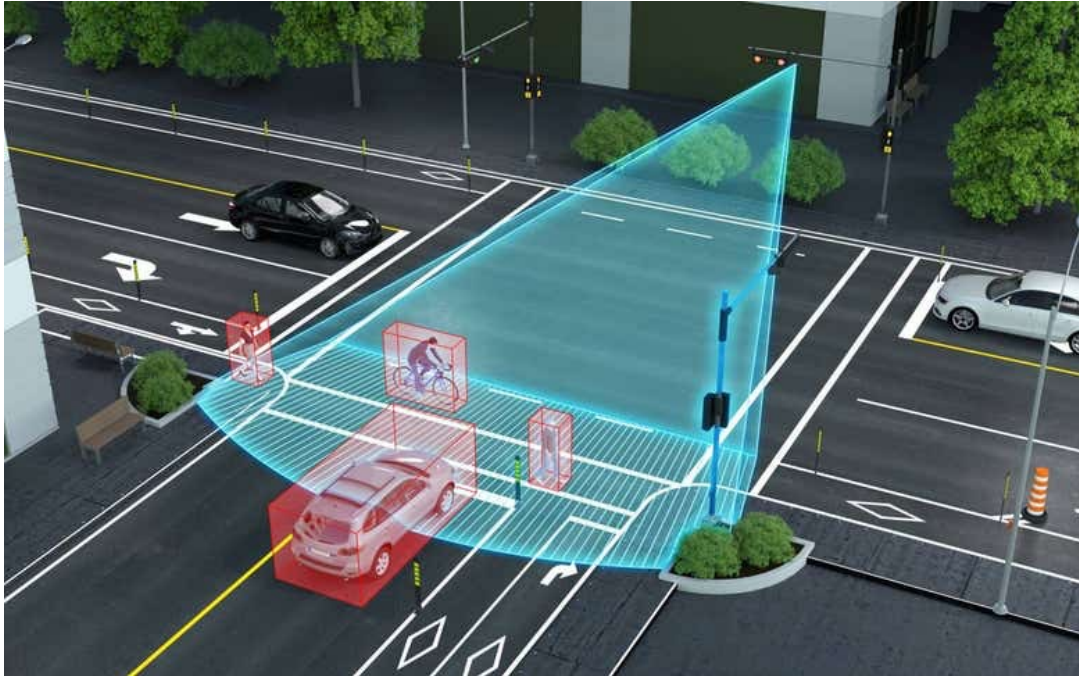
Car Color Detection



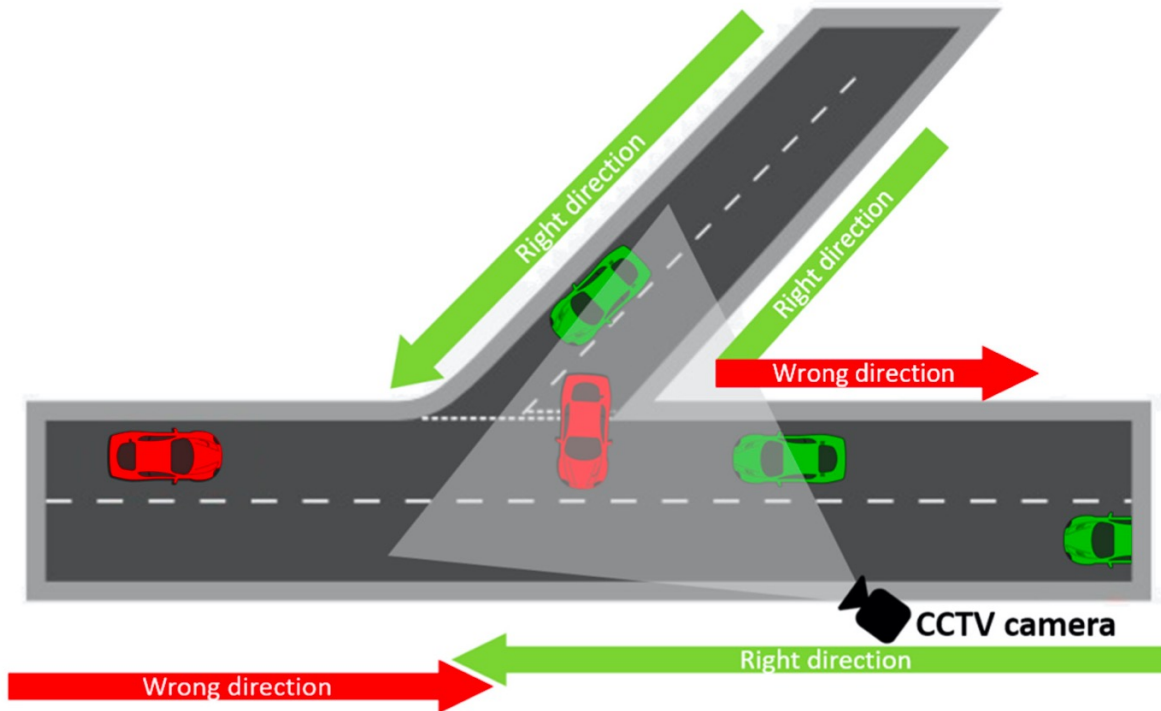
Car Model Detection & Classification



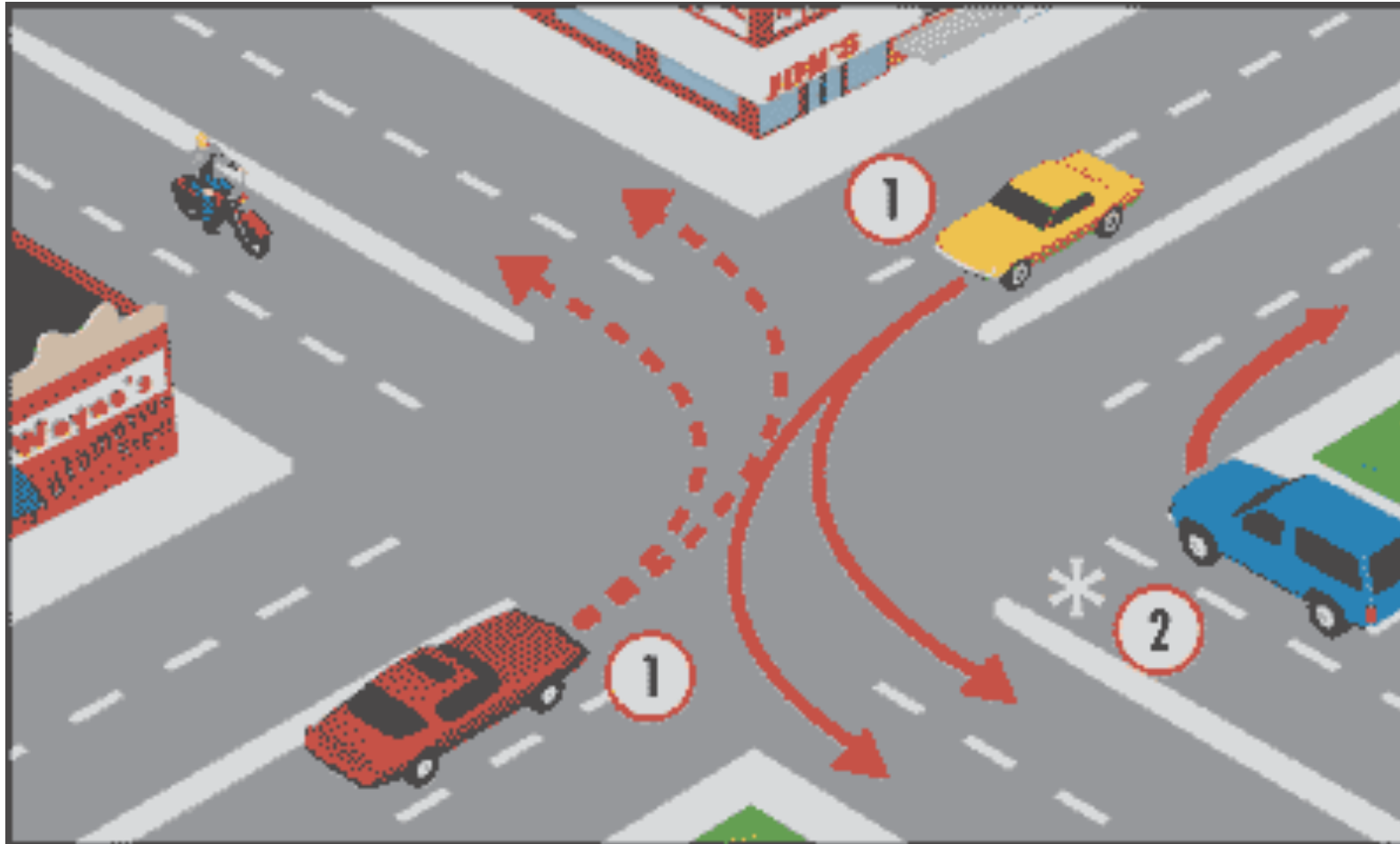
Traffic Violation Detection



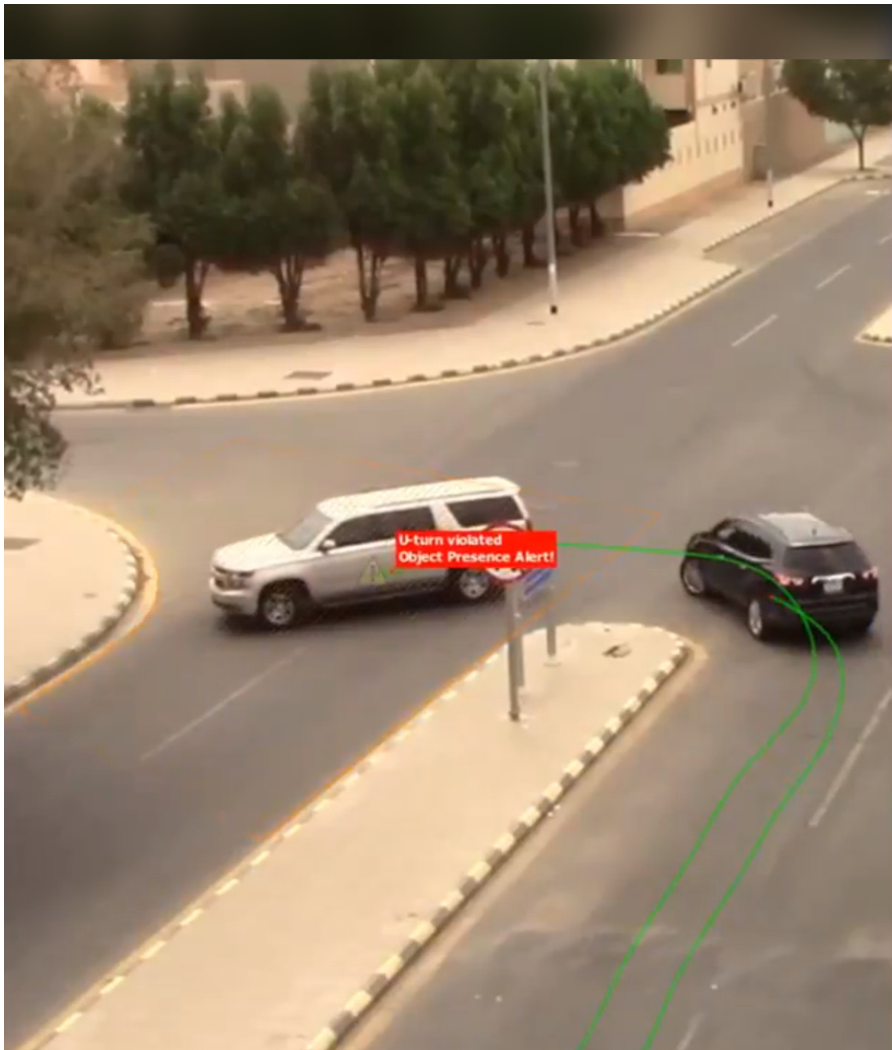
Wrong Direction Detection



Phát hiện xe rẽ trái / phải



Phát hiện xe quay đầu (chữ U)



DATA FROM SKY

#How-to series with FLOW
How-to detect #TrafficViolations #U-turns

Make a **U-turn detector** right now!

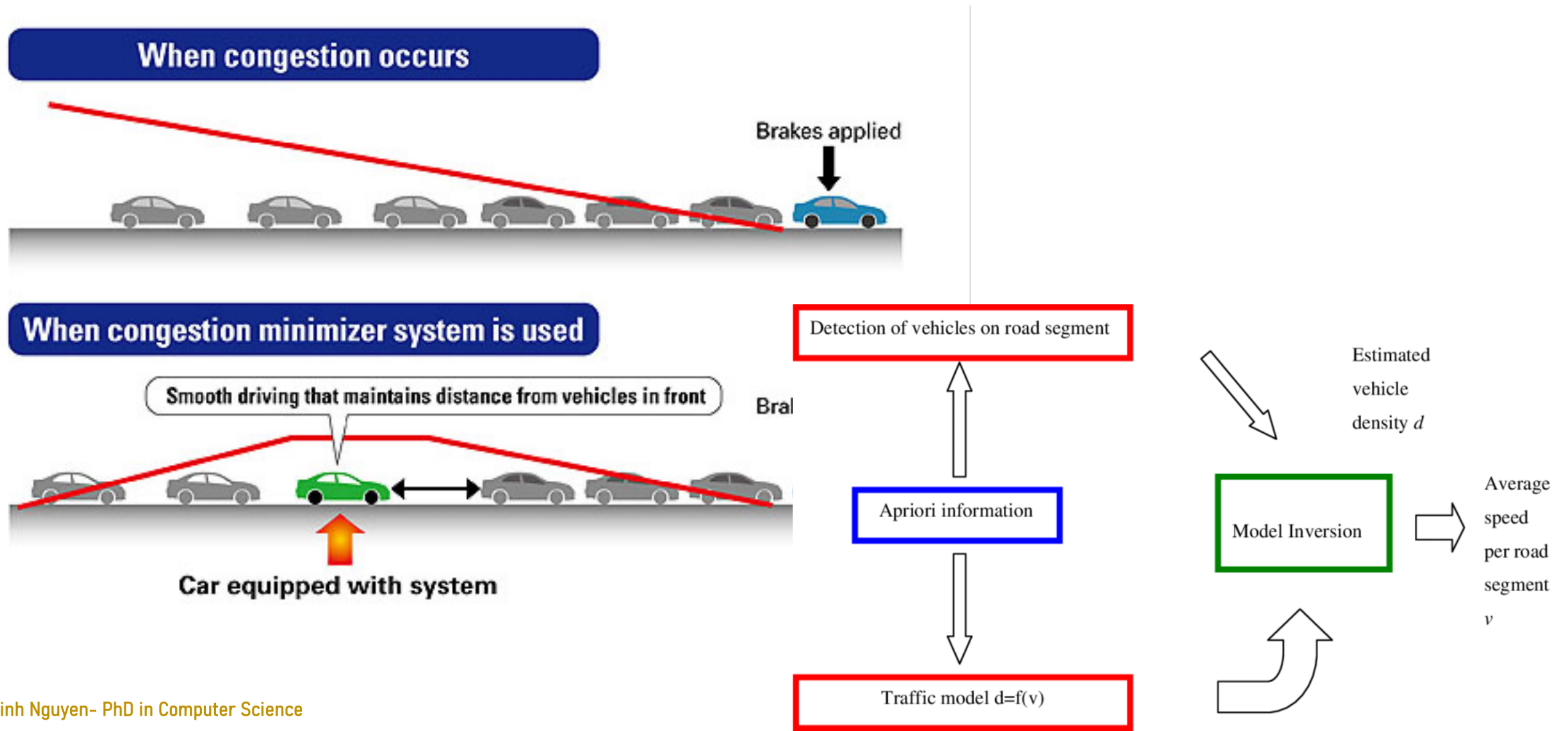
See how to:

- detect **U-turns**
- **visualize** the data on dashboard

datafromsky.com

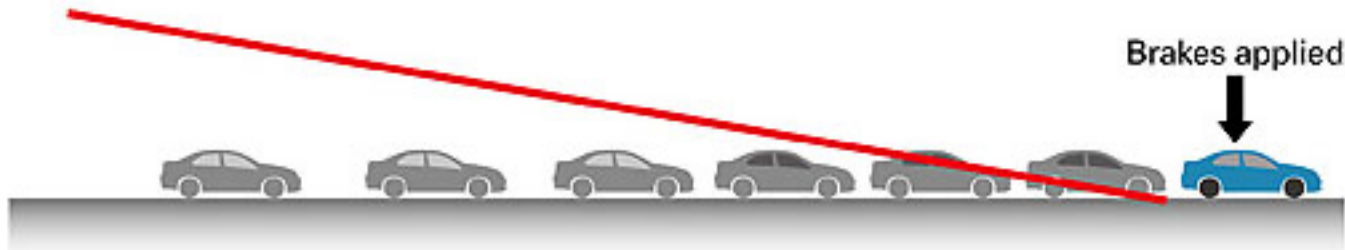
FLOW.
INSIGHTS

Phát hiện ùn tắc giao thông

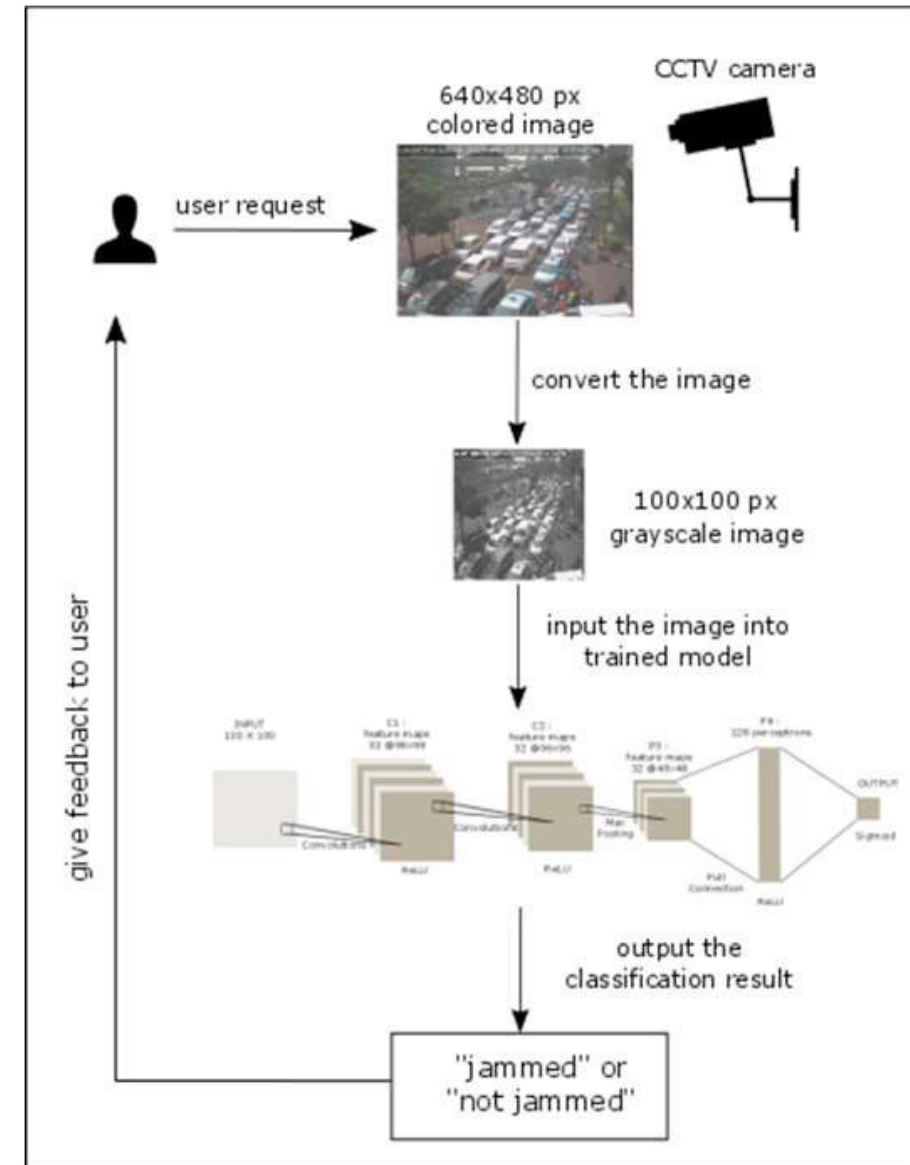
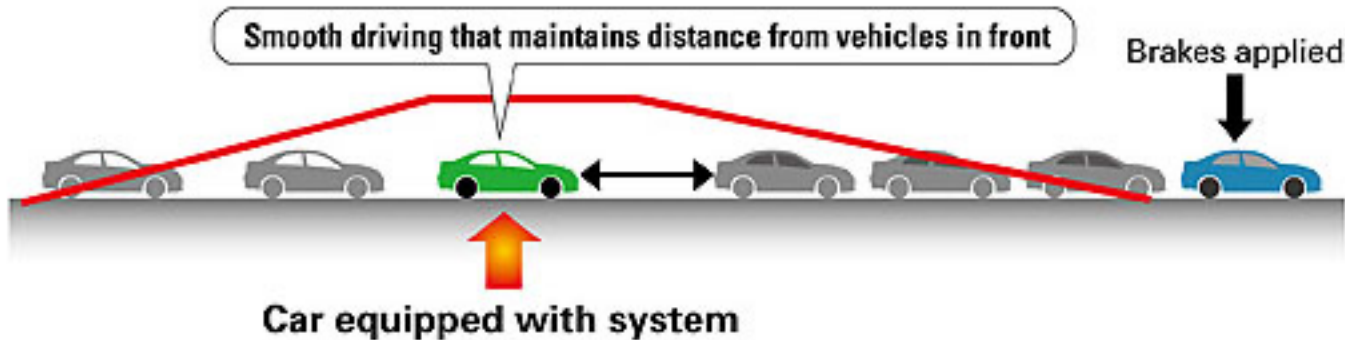


Phát hiện ùn tắc giao thông

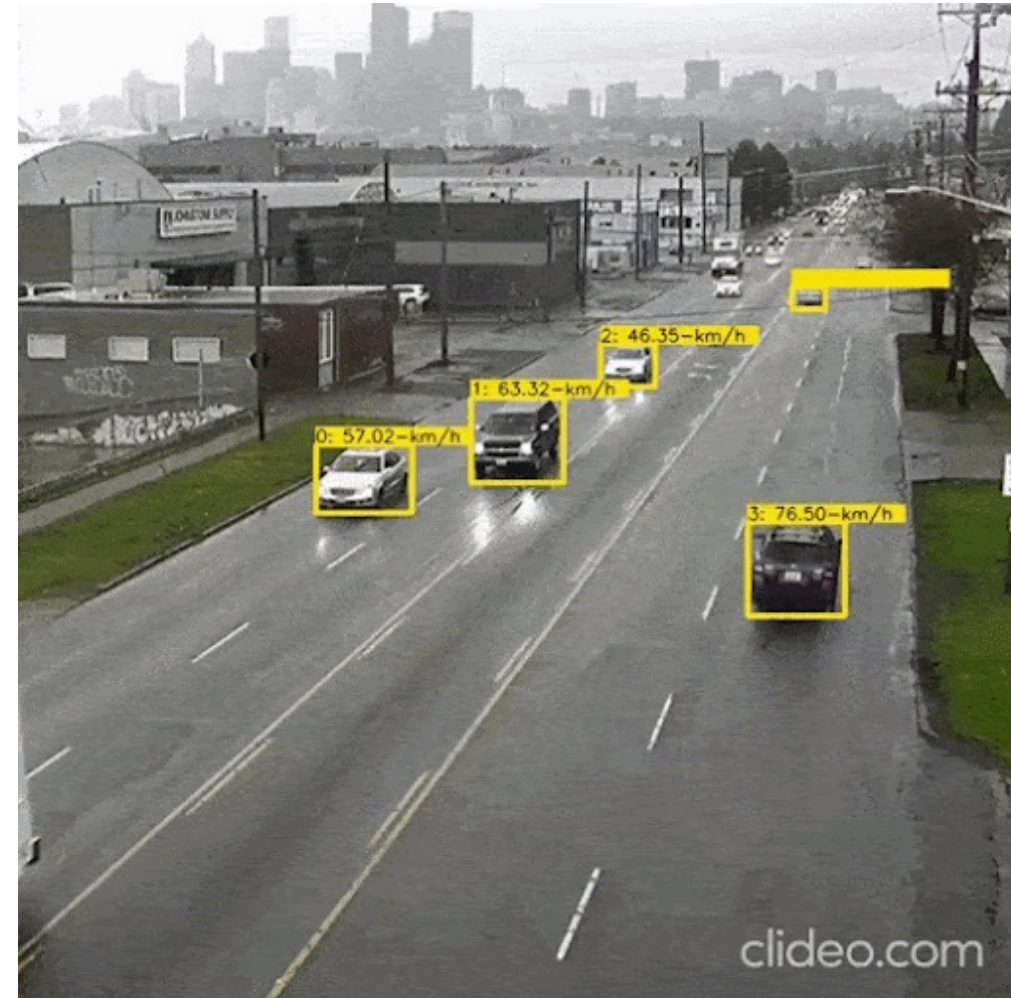
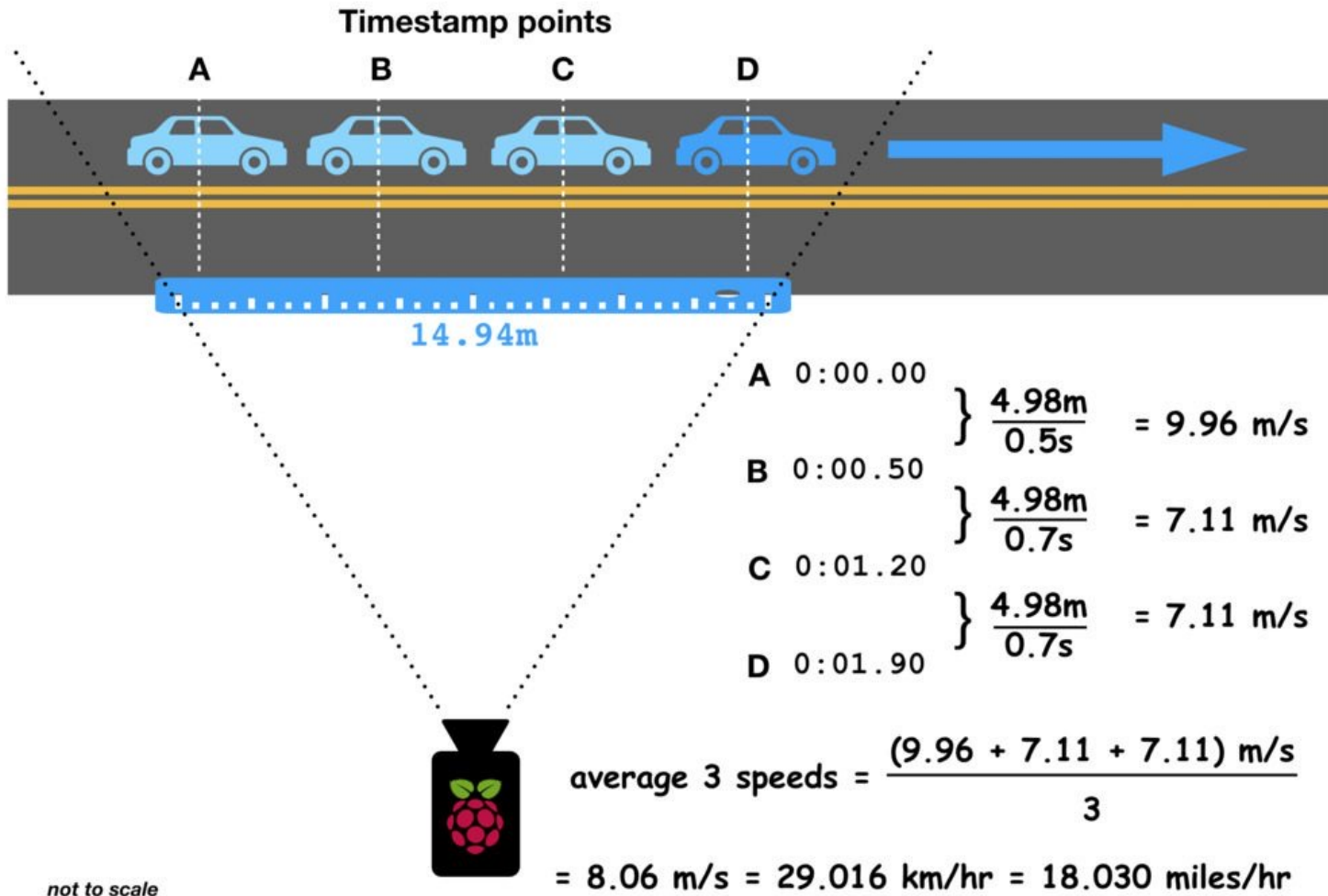
When congestion occurs



When congestion minimizer system is used



Phát hiện xe chạy vượt/dưới tốc độ



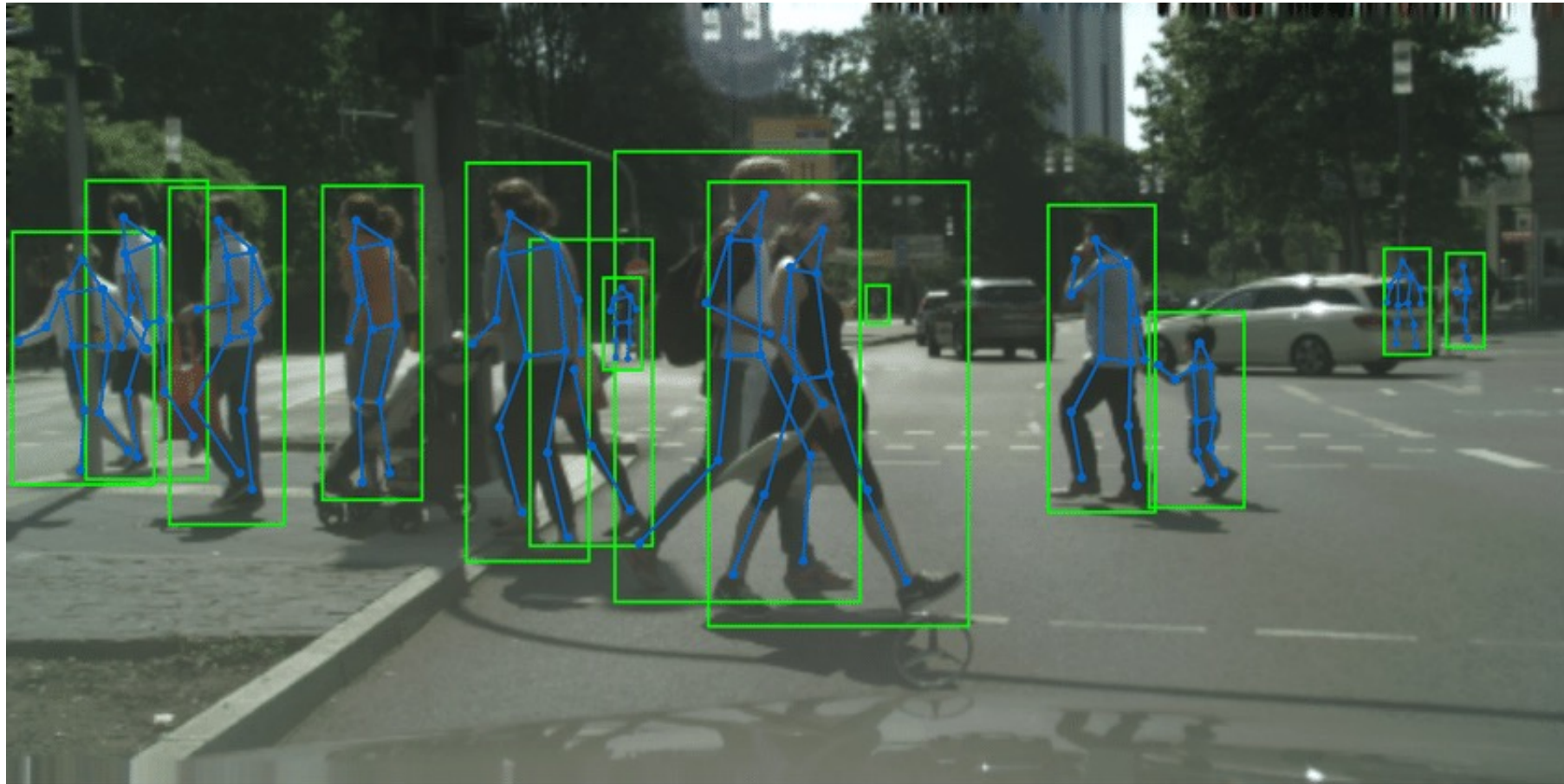
clideo.com

Phát hiện xe dừng trên đường

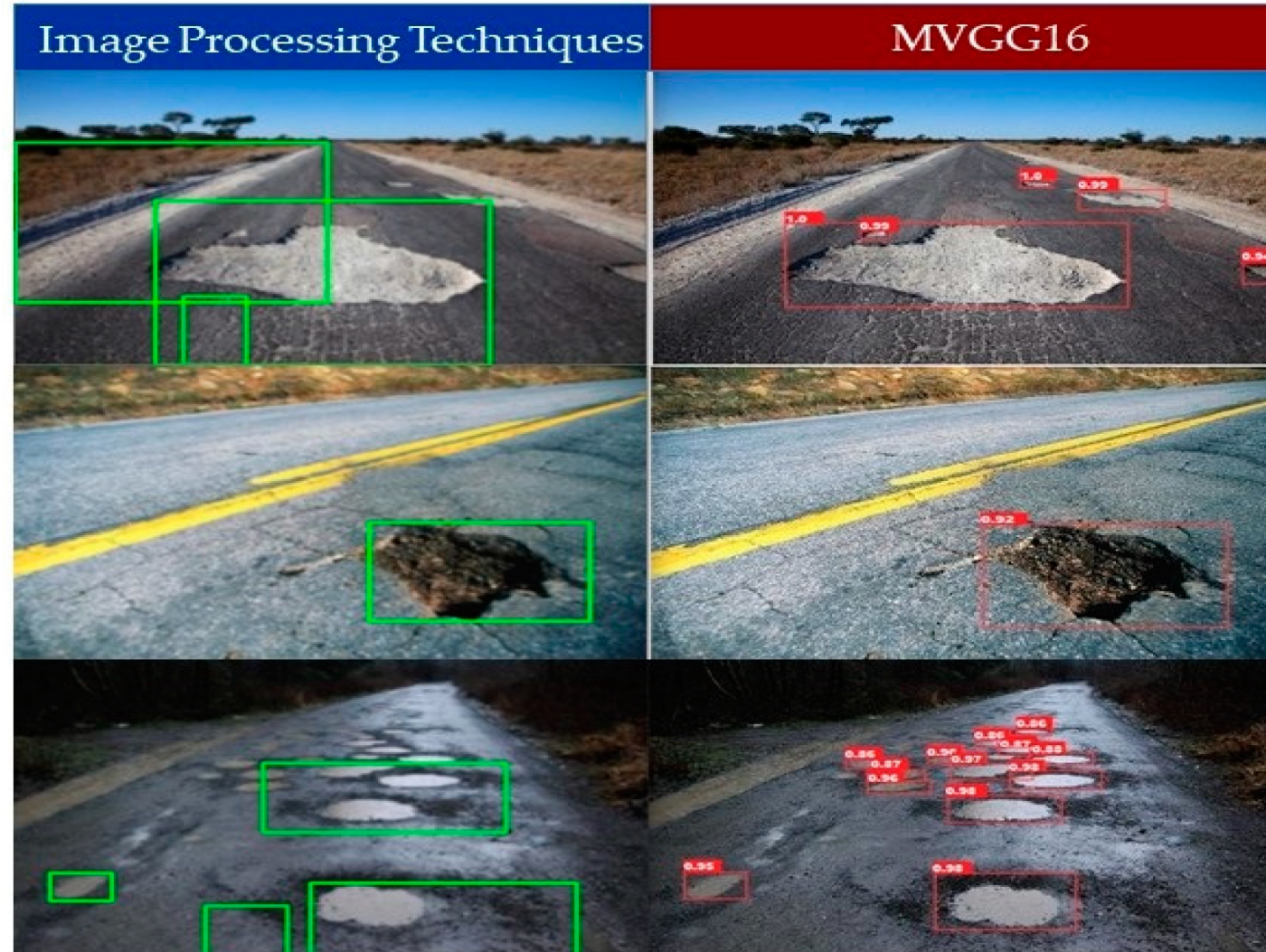
- <https://youtu.be/dLtmLN2Ybos>



Phát hiện người đi bộ



Phát hiện đường hỏng



Phát hiện khói --> tầm nhìn kém



ation



Classification + Localization



Object Detection



Instances



Outline

- **What is Research Topics**
- **How to Find a Good Research Topic**
- **How to Develop Research Questions**
- **Sample Research Topics in Computer Vision**
- **Assignment**

Assignment 2

- **Requirement:**
 - Find and read five papers related to your research interest
 - Find and write down 3-5 Research Questions related to your research interest
 - **Submit to:** aivnresearch@gmail.com
- **Deadline:** 12:00 25/11/2023
- **Receiving Feedback:** 1 week from TA Research Team of AIVN

Assignment 2

AI VIET NAM – COURSE 2023

How to find a Research Topic

(Research Module - Assignment 2)

Ngày 17 tháng 11 năm 2023

Ở bài tập này, đầu tiên các bạn cần tìm kiếm được 5 bài báo nghiên cứu khoa học dựa trên research interest của mình (Ưu tiên tìm đọc các bài survey được viết bởi các tác giả đầu ngành). Sau đó các bạn tiến hành đọc bài báo và hoàn thành câu trả lời cho các câu hỏi bên dưới.

Lưu ý: sau khi hoàn thành bài tập và submit bài tập này, các bạn sẽ nhận được feedback từ đội ngũ TA research của AIVN trong vòng 1 tuần nhé!

Câu hỏi 1 Thông tin bài báo survey/review:

(a) What is the title of the survey/review paper?

Answer :

,

,

(b) What are the remaining challenges related to a specific topic mentioned in a paper?

Answer :

,

,

